03

时序逻辑电路

3 - 1 锁存器、触发器

3 - 2 计数器

3 - 3 时序逻辑电路设计

1. 按逻辑功能划分有：

计数器、寄存器、读/写存储器、顺序脉冲发生器等。

2. 根据存储电路中触发器的动作特点不同分为：

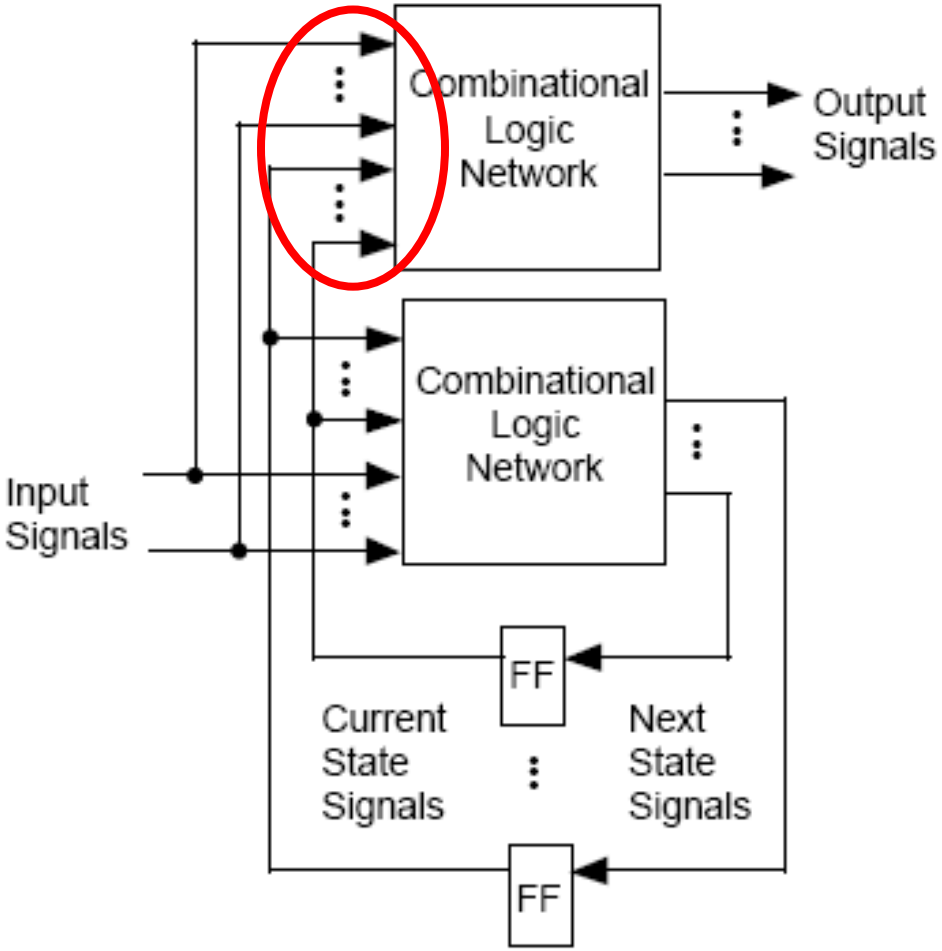
同步时序电路和异步时序电路。

synchronous and asynchronous

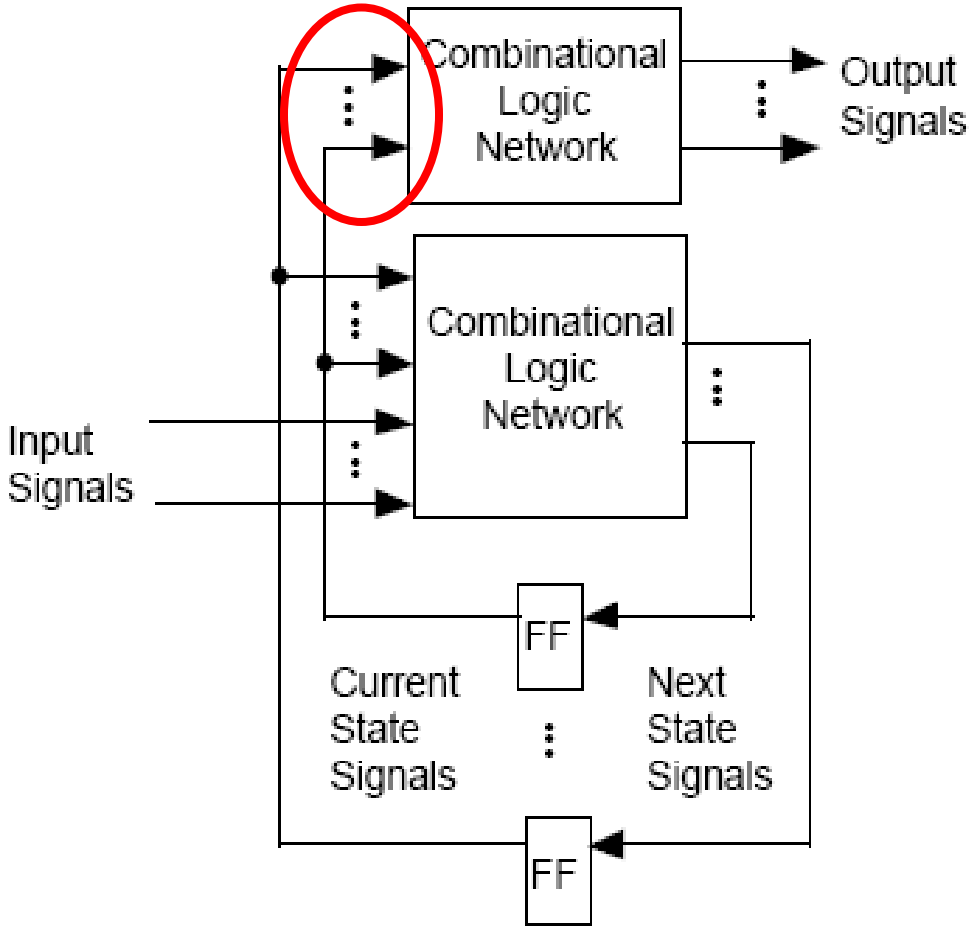
3. 根据输出信号的特点分为：

Mealy Model和Moore Model 。

Mealy Model: *Outputs depend on current state and inputs*



Moore Model: *Outputs only depend on current state.*



计数器 Counter

1）计数器的作用

记忆输入脉冲的个数；用于计数、定时、分频、产生节拍脉冲及进行数字运算等。

2）计数器的分类

按计数脉冲引入的方式分：同步计数器和异步计数器。

按数字增减的方式分：加法、减法和可逆计数器。

按数字编码的方式分：如二进制计数器、二－十进制计数器等。

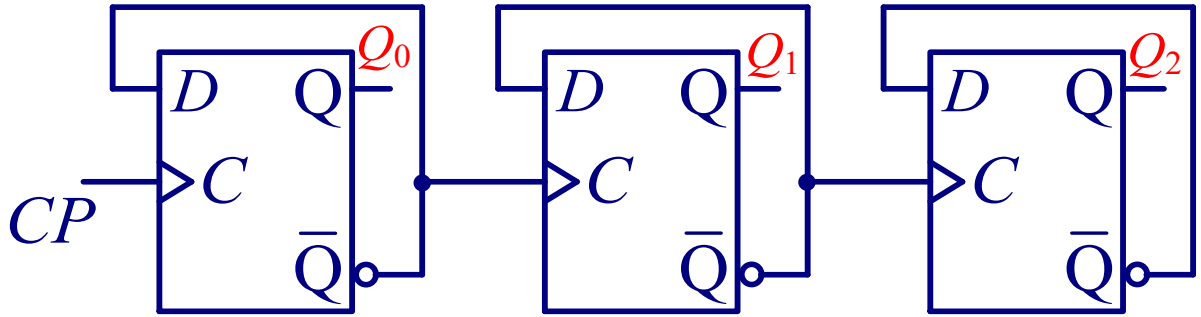
按计数器的容量分：如十进制、十六进制。

3）计数器的构成　JK或D触发器、集成计数器

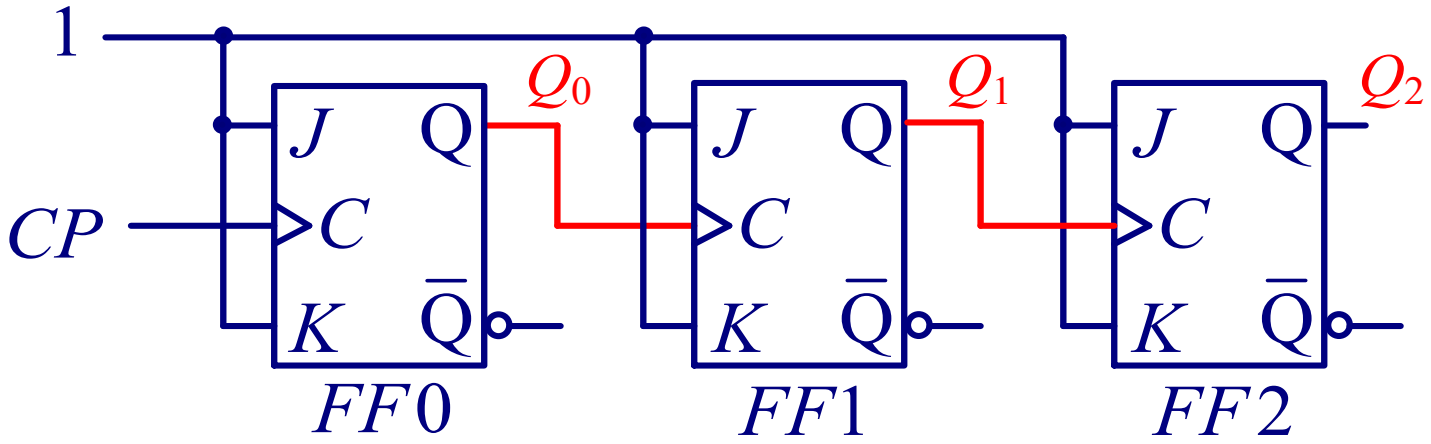
1、异步计数器 Asynchronous Counter

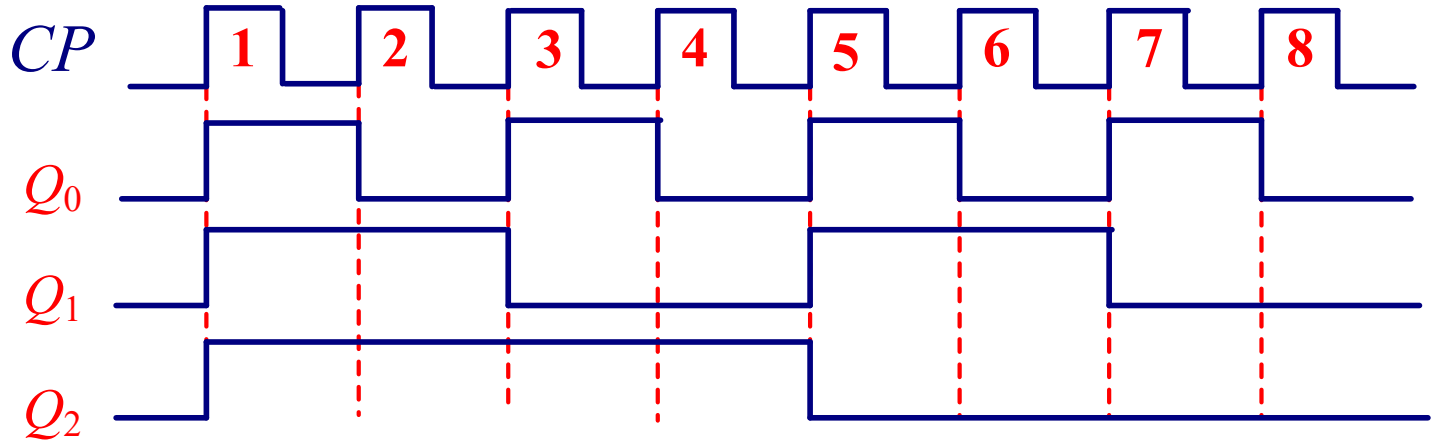
1.1 异步二进制计数器 ripple counter

特点：异步计数器内部，有的触发器直接受输入计数脉冲控制，有的触发器则是把其它触发器的输出信号作为自己的时钟脉冲，因此各个触发器状态变换的时间先后不一，故被称为“异步计数器”。

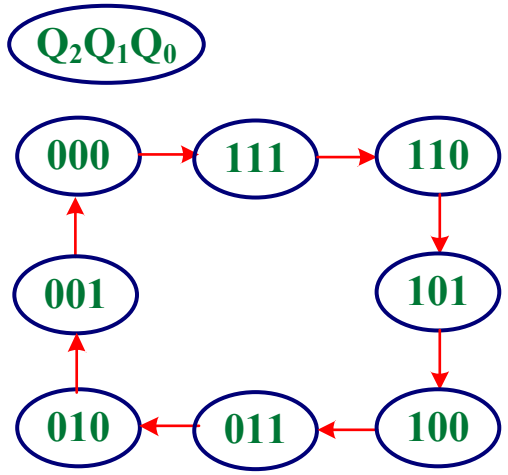


Asynchronous Binary Counter



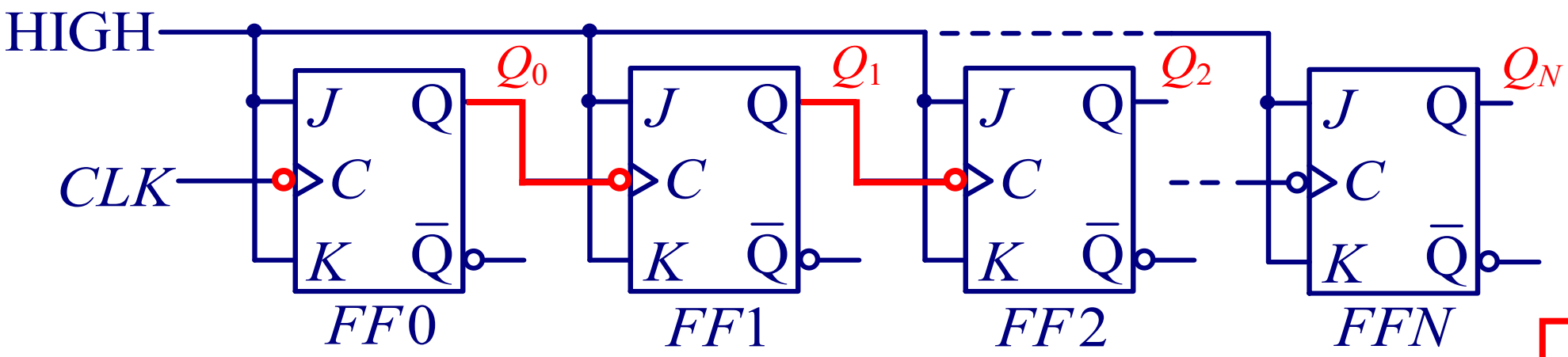


CP	Q_2	Q_1	Q_0
0	0	0	0
1	1	1	1
2	1	1	0
3	1	0	1
4	1	0	0
5	0	1	1
6	0	1	0
7	0	0	1
8	0	0	0

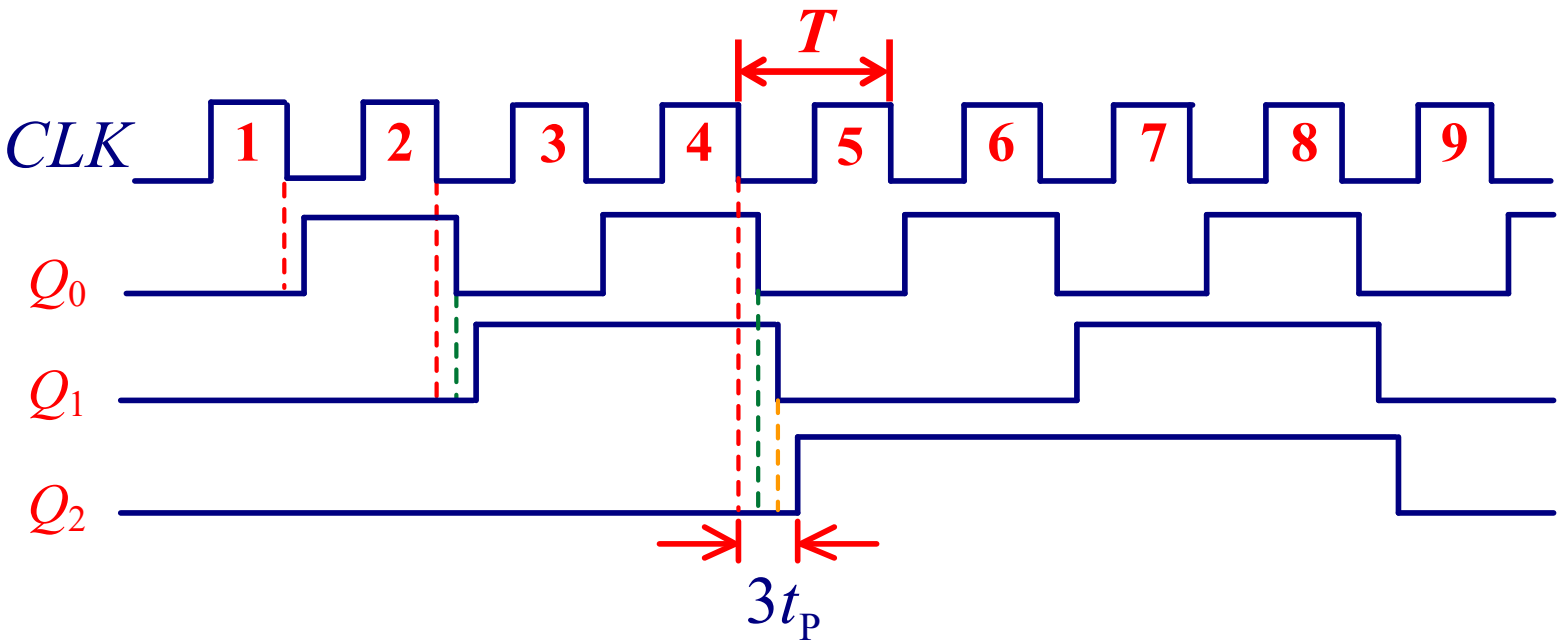


三位二进制异步减法计数器

Asynchronous Binary Counter



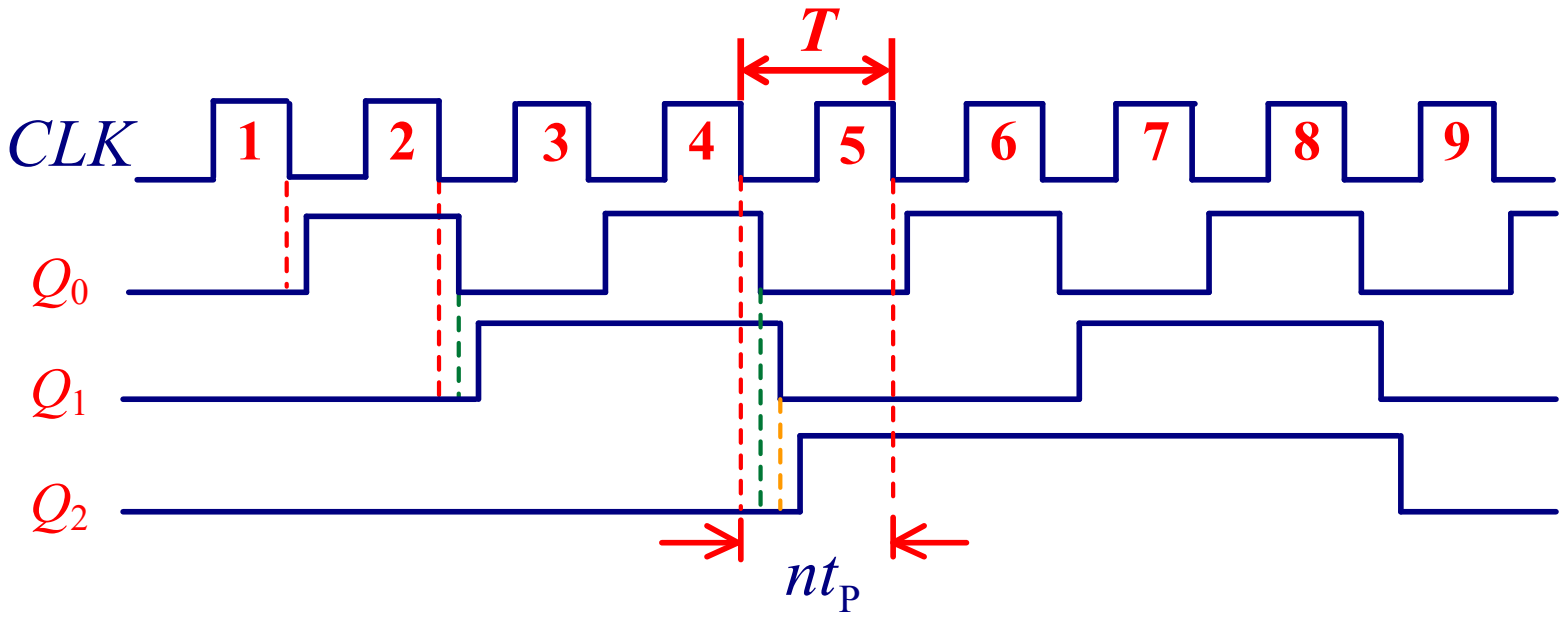
2^n



n是否有限制?

$$T \geq nt_P$$

$$n \leq \frac{T}{t_P}$$



$$T_{min} \geq nt_P$$

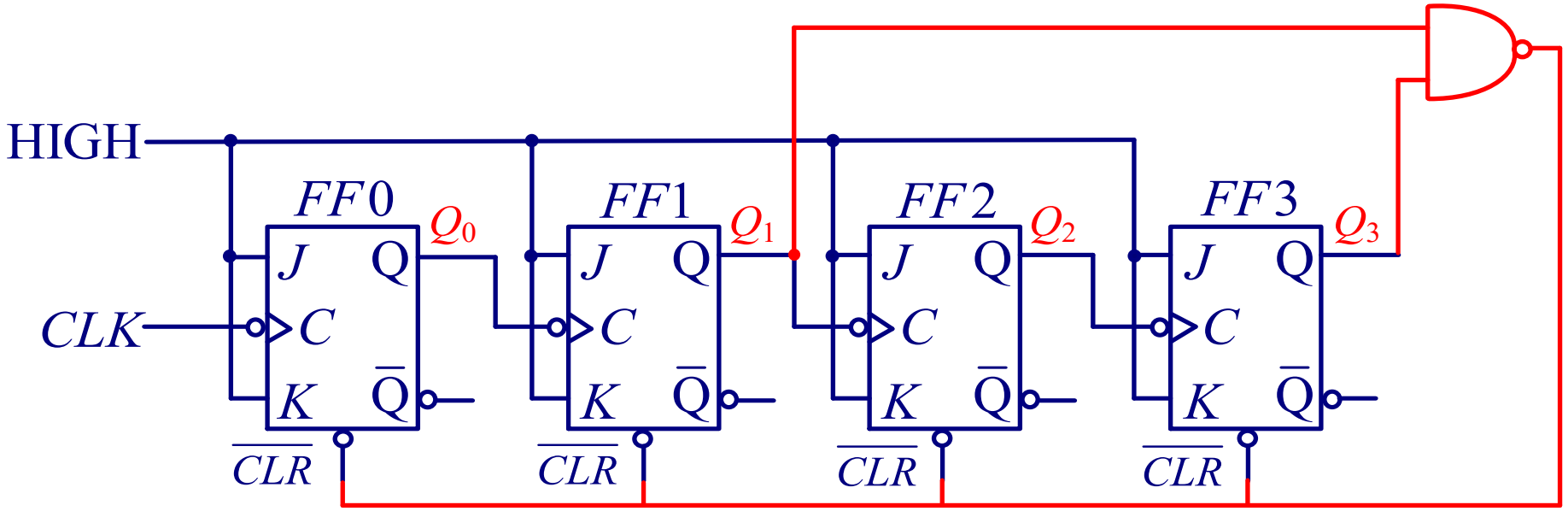
$$f_{max} \leq \frac{1}{nt_P}$$

异步计数器的时钟频率受触发器的传输延迟限制

例：若构成3位异步计数器的JK边沿触发器的传输延迟为10ns，求该计数器所允许的最大时钟频率。

$$T_{min} = 3t_p = 30ns \qquad f_{max} = \frac{1}{T_{min}} = 33.3MHz$$

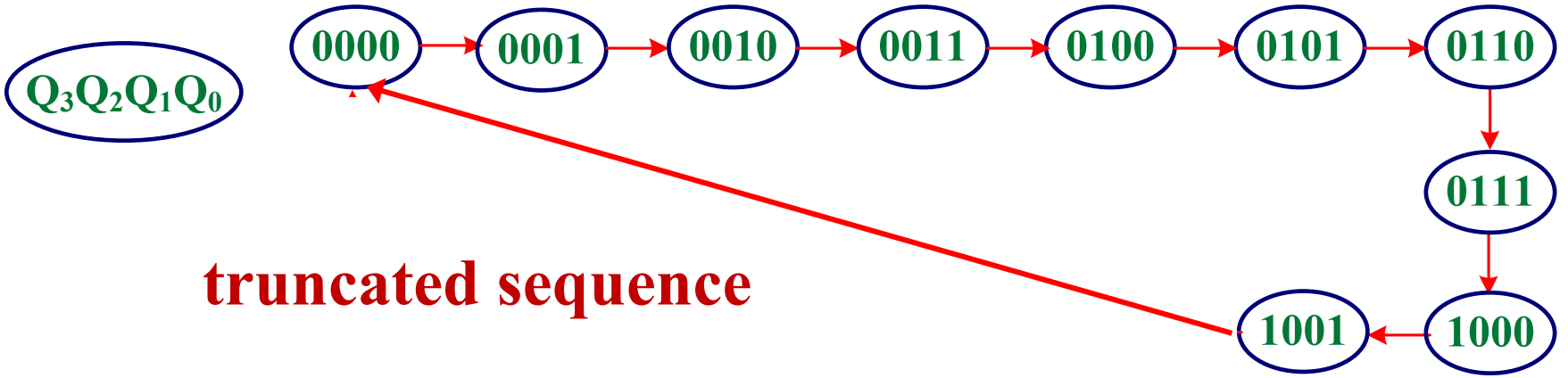
1.2 异步十进制计数器 Asynchronous Decade Counters



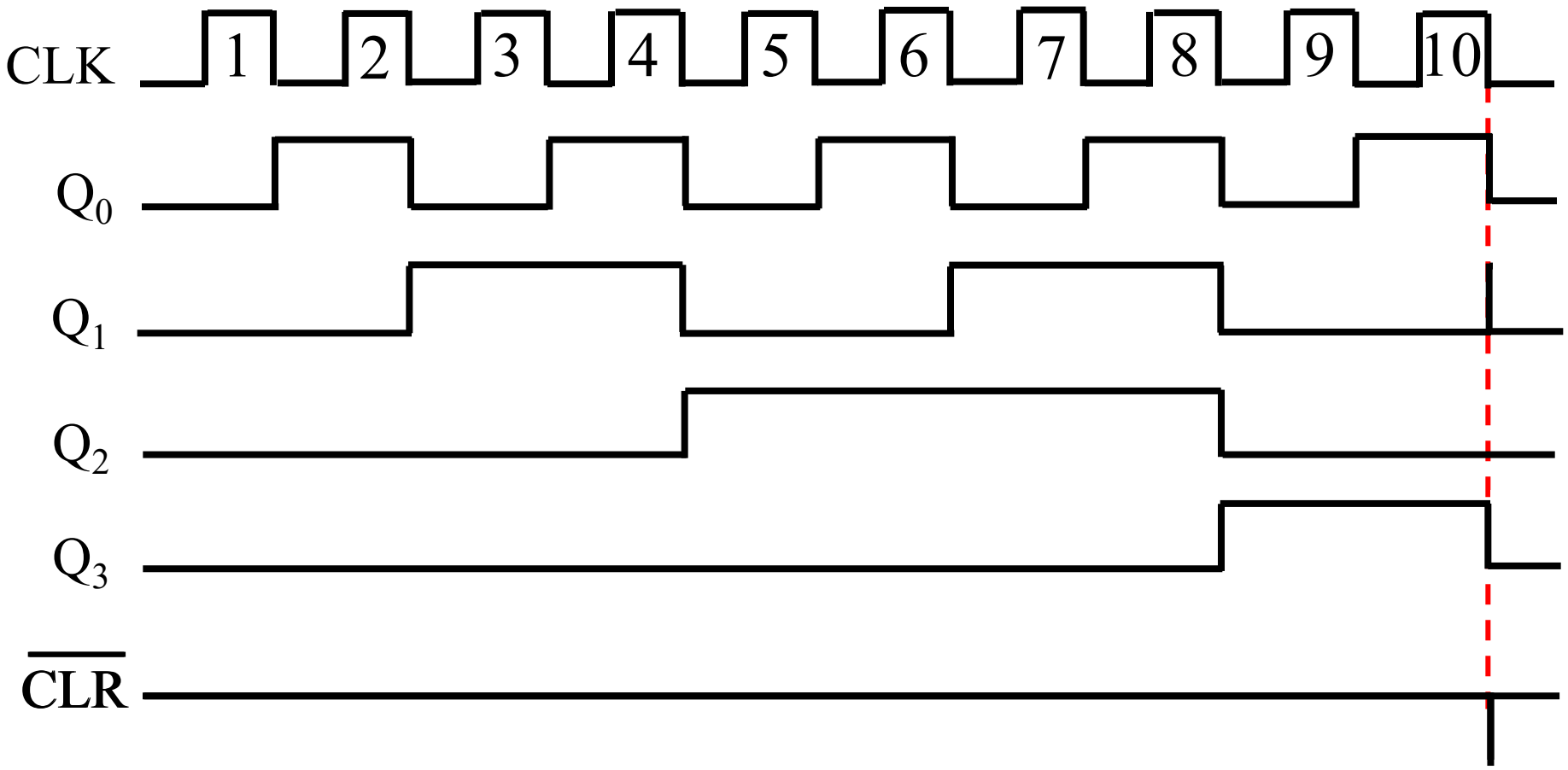
Partial decoding

$n \rightarrow 2^n$

module

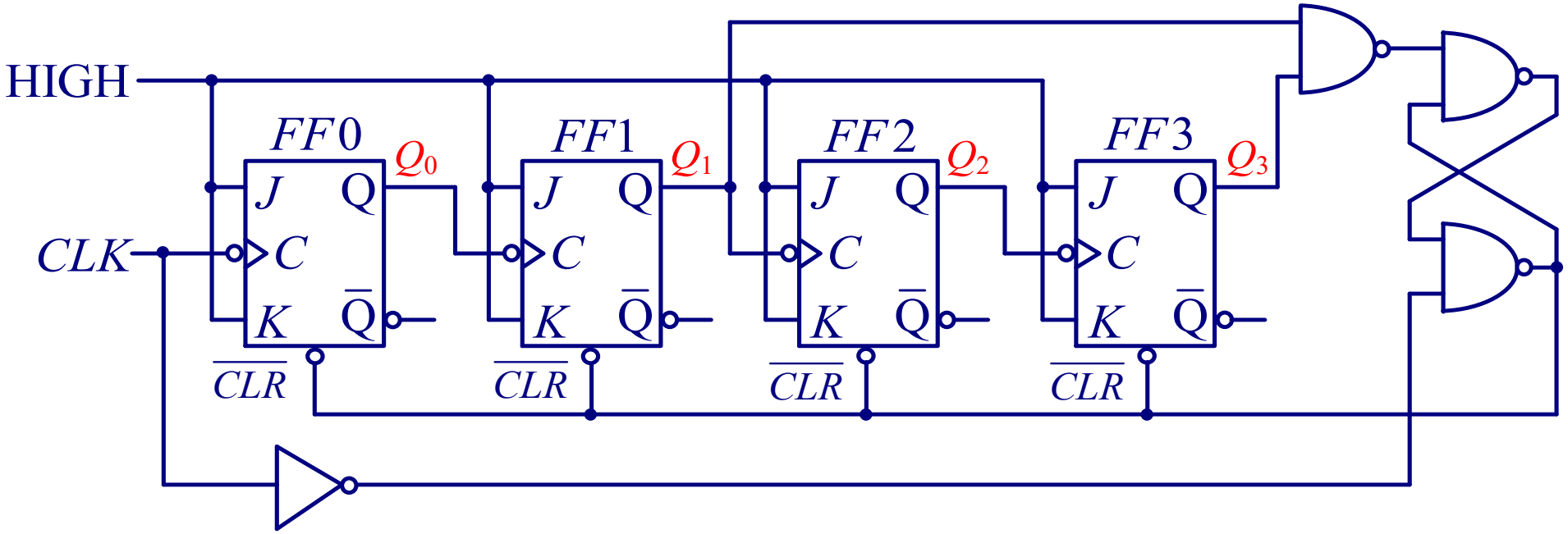


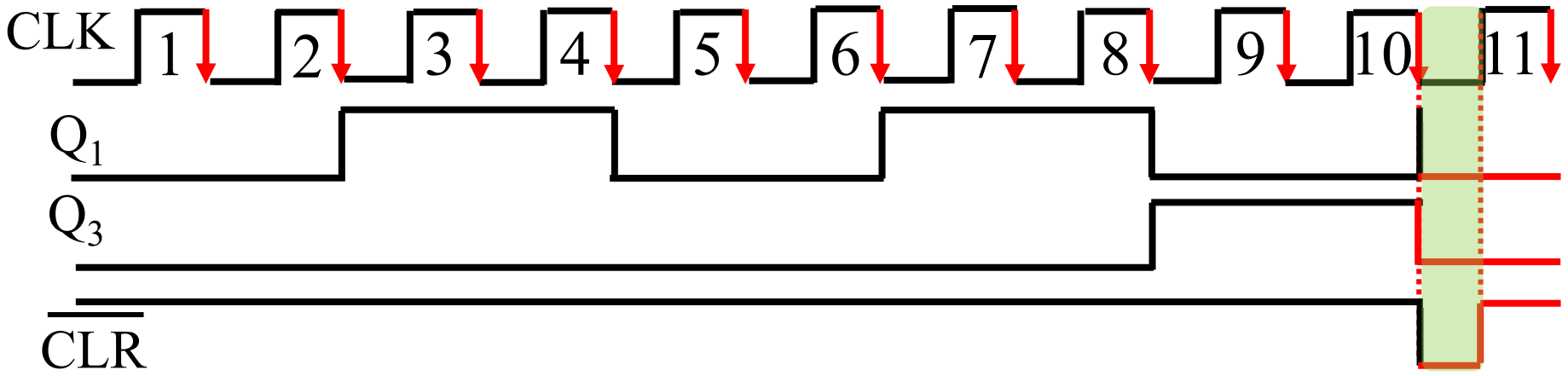
truncated sequence



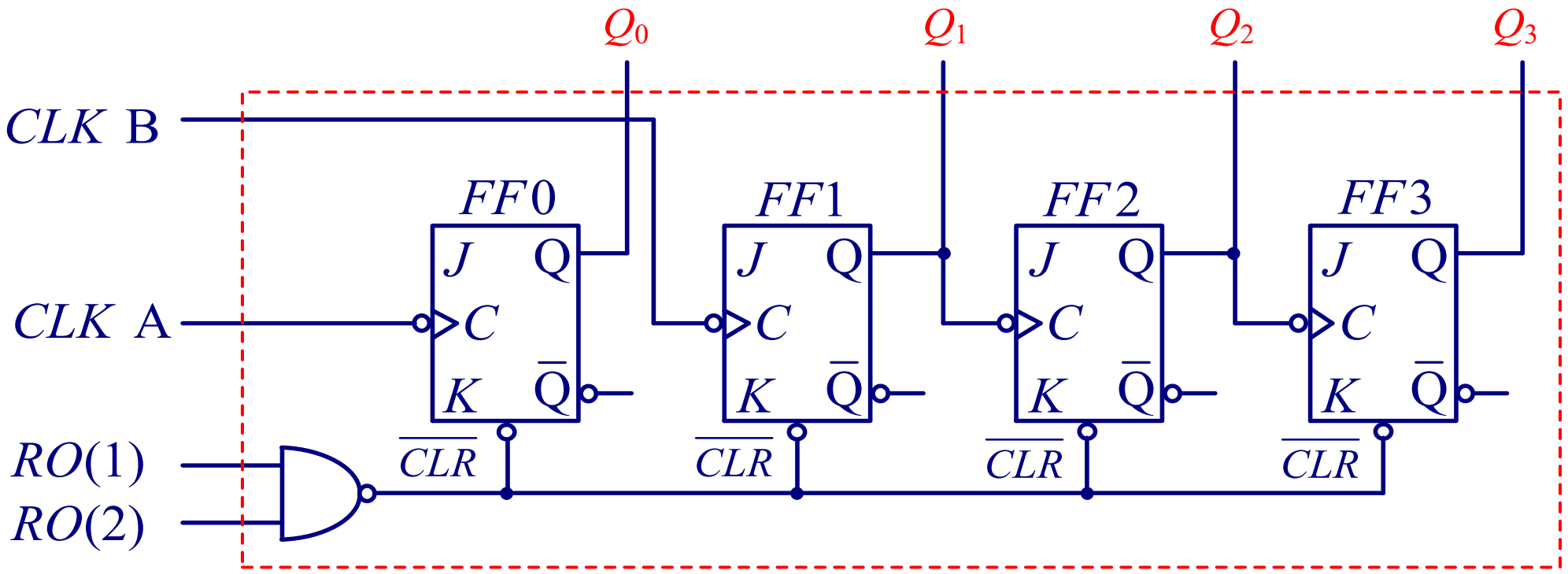
glitch: a false or spurious electronic signal.

缺点：置零信号持续时间极短，动作慢的触发器可能还未复位，置零信号就已消失，故电路的可靠性不高


HIGH
CLK
FF0
FF1
FF2
FF3
J
C
K
Q
$\overline{Q}$
Q_0
Q_1
Q_2
Q_3
$\overline{CLR}$


CLK
1
2
3
4
5
6
7
8
9
10
11
Q_1
Q_3
$\overline{CLR}$

74LS93：A 4-bit Asynchronous Binary Counter

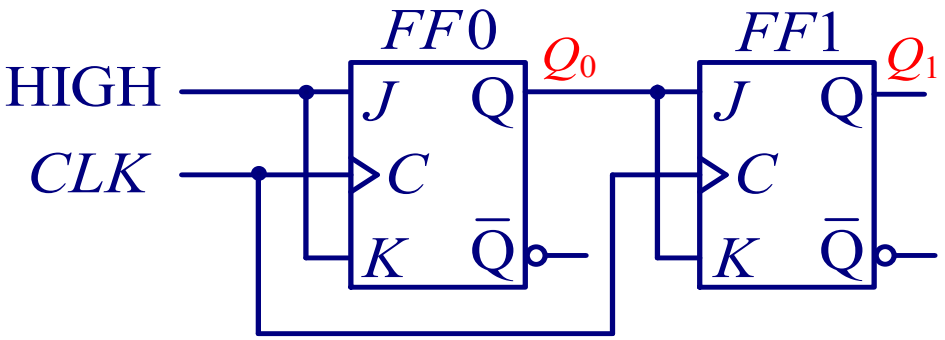


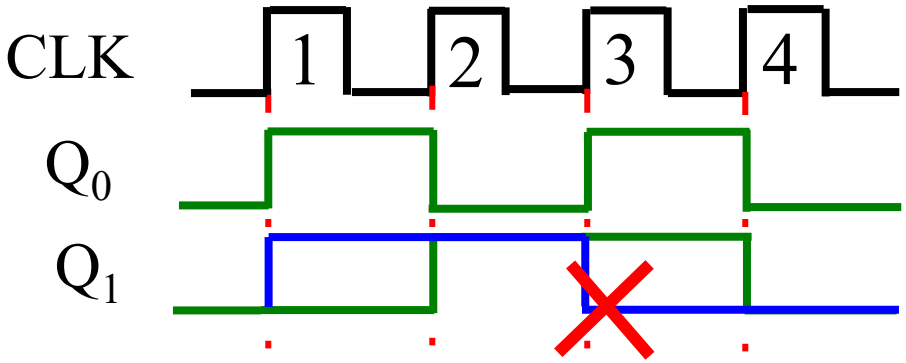
门控复位输入，可以用来实现模小于16的任意计数器

2、同步计数器 Synchronous Counter

特点：在同步计数器内部，各个触发器都受同一时钟脉冲——输入计数脉冲的控制，因此，它们状态的更新几乎是同时的，故被称为“同步计数器”。

例：两位二进制同步加法计数器。

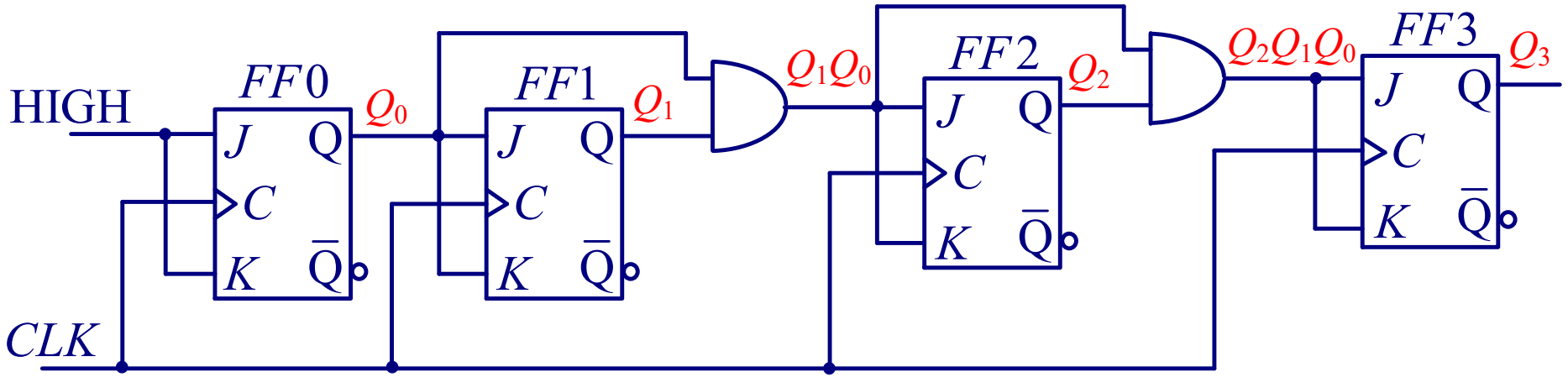


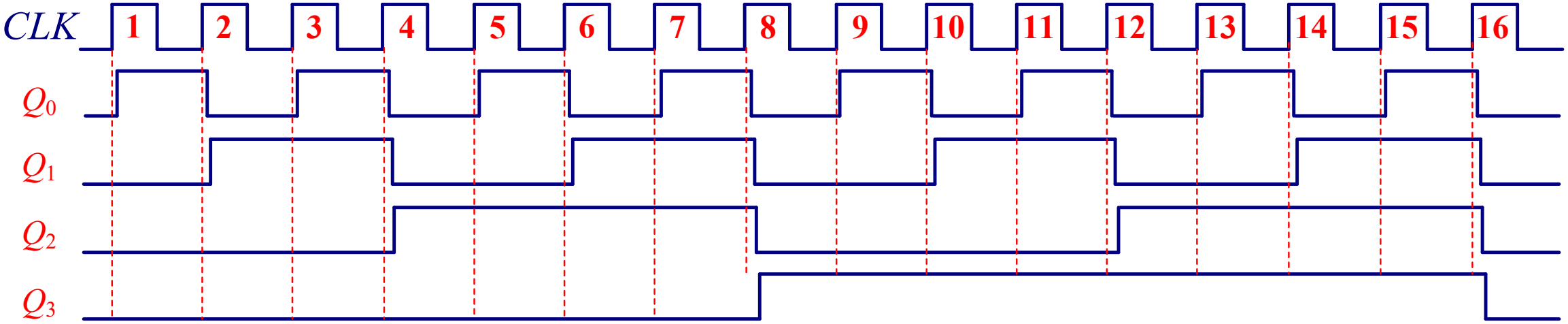


时序图中忽略传输延迟时

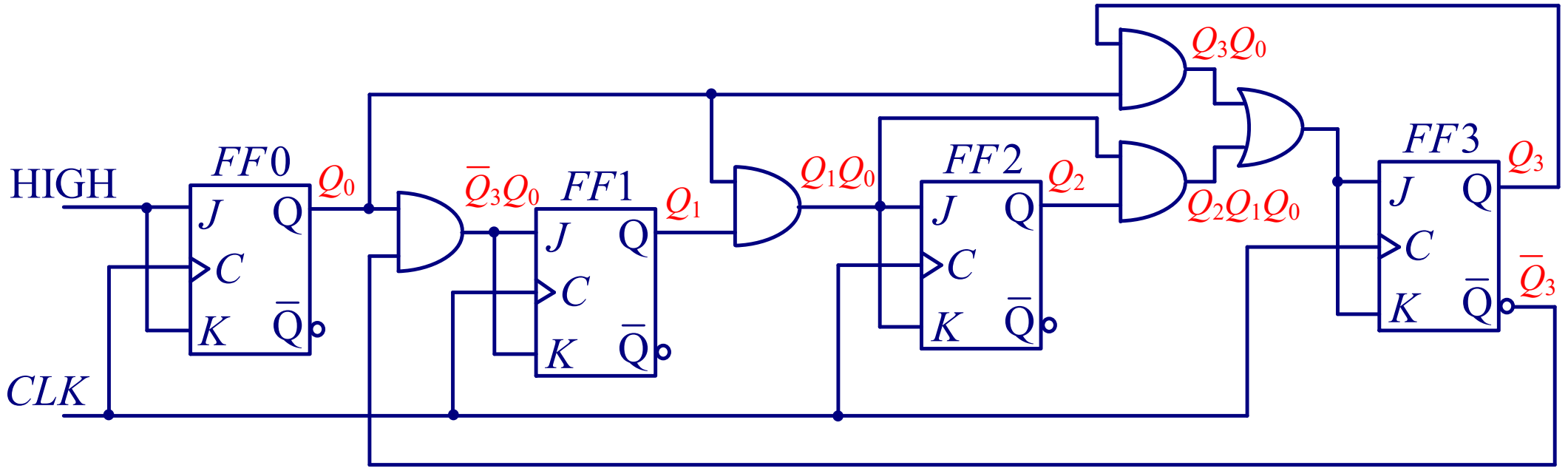
传输延迟是同步计数器运算的重要因素

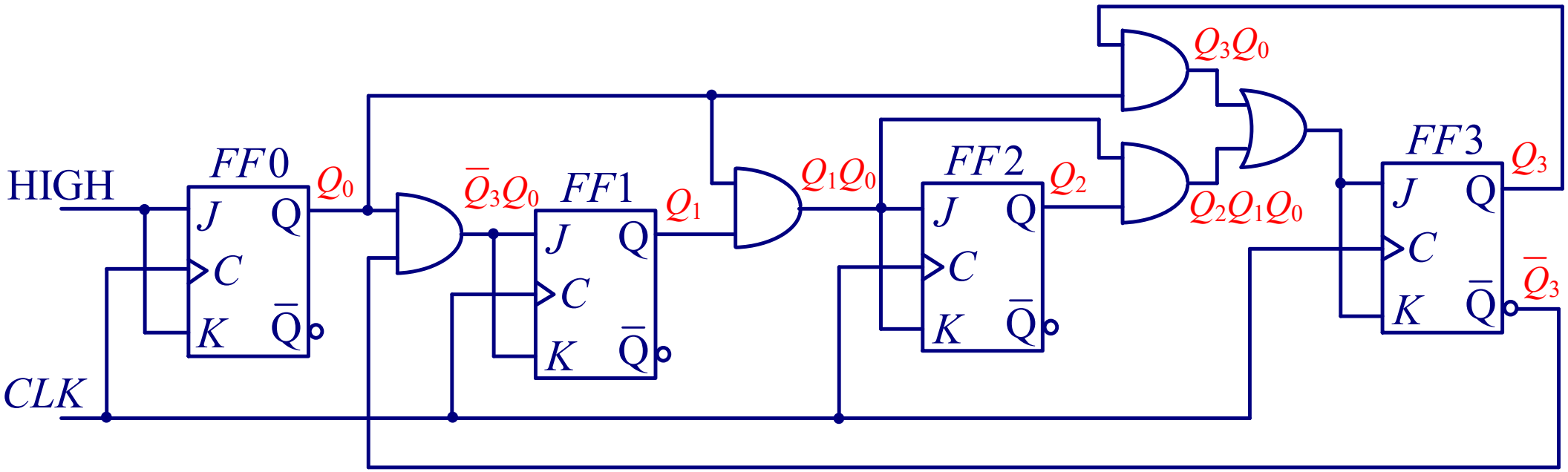
2.1 同步二进制计数器 A 4-bit Synchronous Binary Counter





2.2 同步十进制计数器 A 4-bit Synchronous Decade Counter





驱动方程 → 特性方程 $Q^{n+1} = J\bar{Q}^n + \bar{K}Q^n$

$J_0 = K_0 = 1$

$J_1 = K_1 = \bar{Q}_3^n Q_0^n$

$J_2 = K_2 = Q_1^n Q_0^n$

$J_3 = K_3 = Q_3^n Q_0^n + Q_2^n Q_1^n Q_0^n$

状态方程

$Q_0^{n+1} = \bar{Q}_0^n$

$Q_1^{n+1} = (\bar{Q}_3^n Q_0^n) \oplus Q_1^n$

$Q_2^{n+1} = (Q_1^n Q_0^n) \oplus Q_2^n$

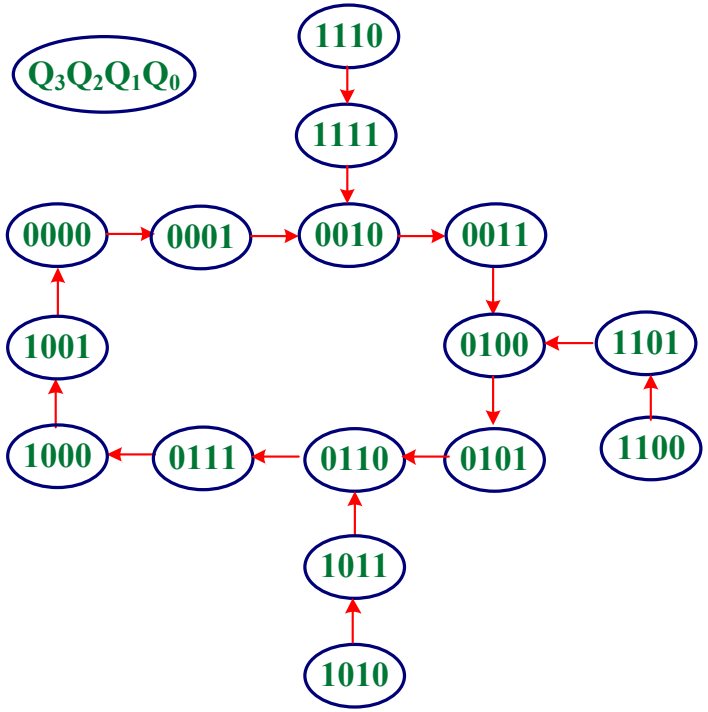
$Q_3^{n+1} = (Q_3^n Q_0^n + Q_2^n Q_1^n Q_0^n) \oplus Q_3^n$

$$Q_0^{n+1}=\bar{Q}_0^n \qquad Q_1^{n+1}=(\bar{Q}_3^n Q_0^n)\oplus Q_1^n \qquad Q_2^{n+1}=(Q_1^n Q_0^n)\oplus Q_2^n \qquad Q_3^{n+1}=(Q_3^n Q_0^n + Q_2^n Q_1^n Q_0^n)\oplus Q_3^n$$

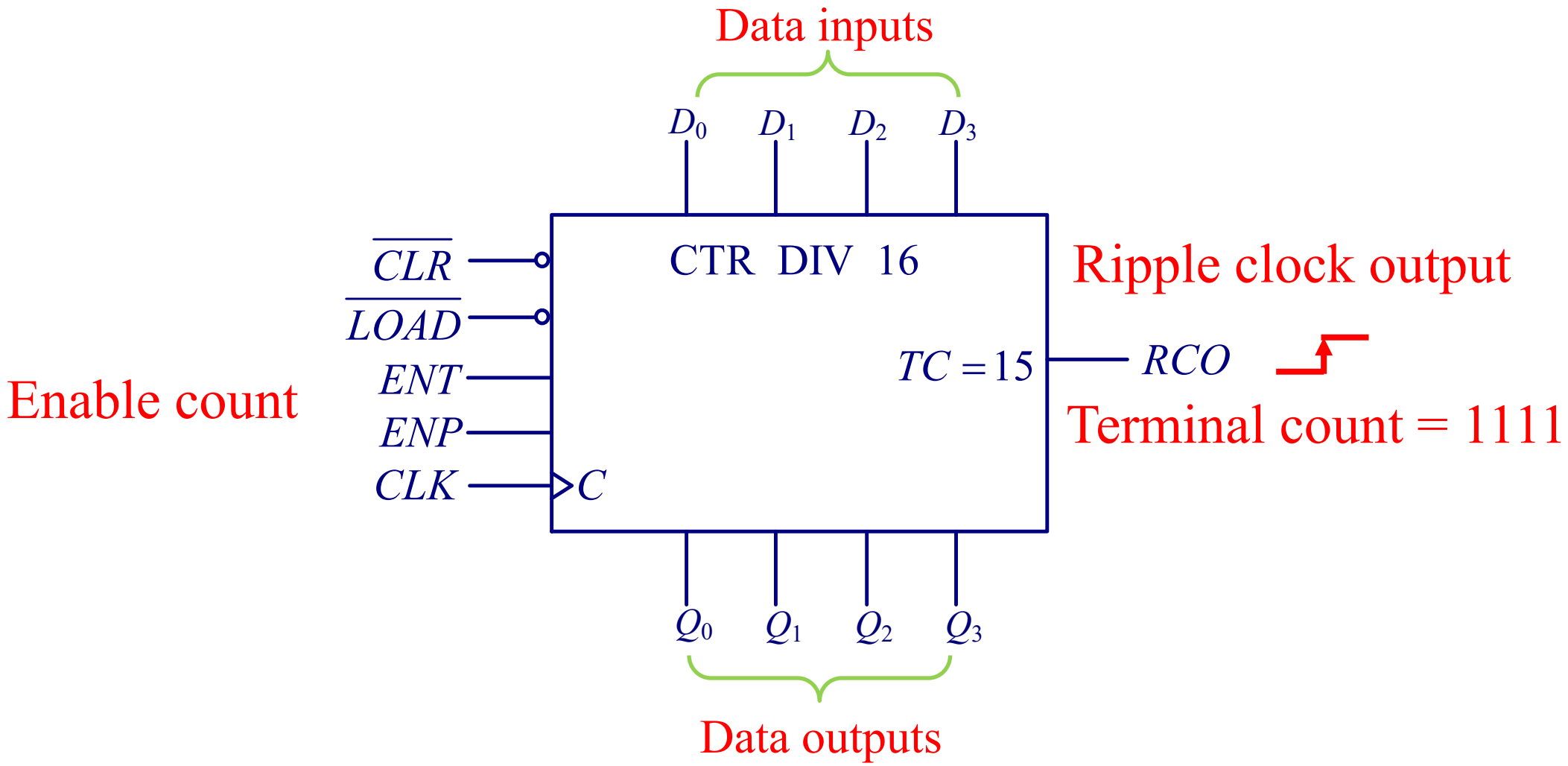
CP	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	0	0	0	0

CP	Q_3	Q_2	Q_1	Q_0
0	1	0	1	0
1	1	0	1	1
2	0	1	1	0
0	1	1	0	0
1	1	1	0	1
2	0	1	0	0
0	1	1	1	0
1	1	1	1	1
2	0	0	1	0

状态转换图：



74HC163：A 4-bit IC Synchronous Binary Counter



Function Table

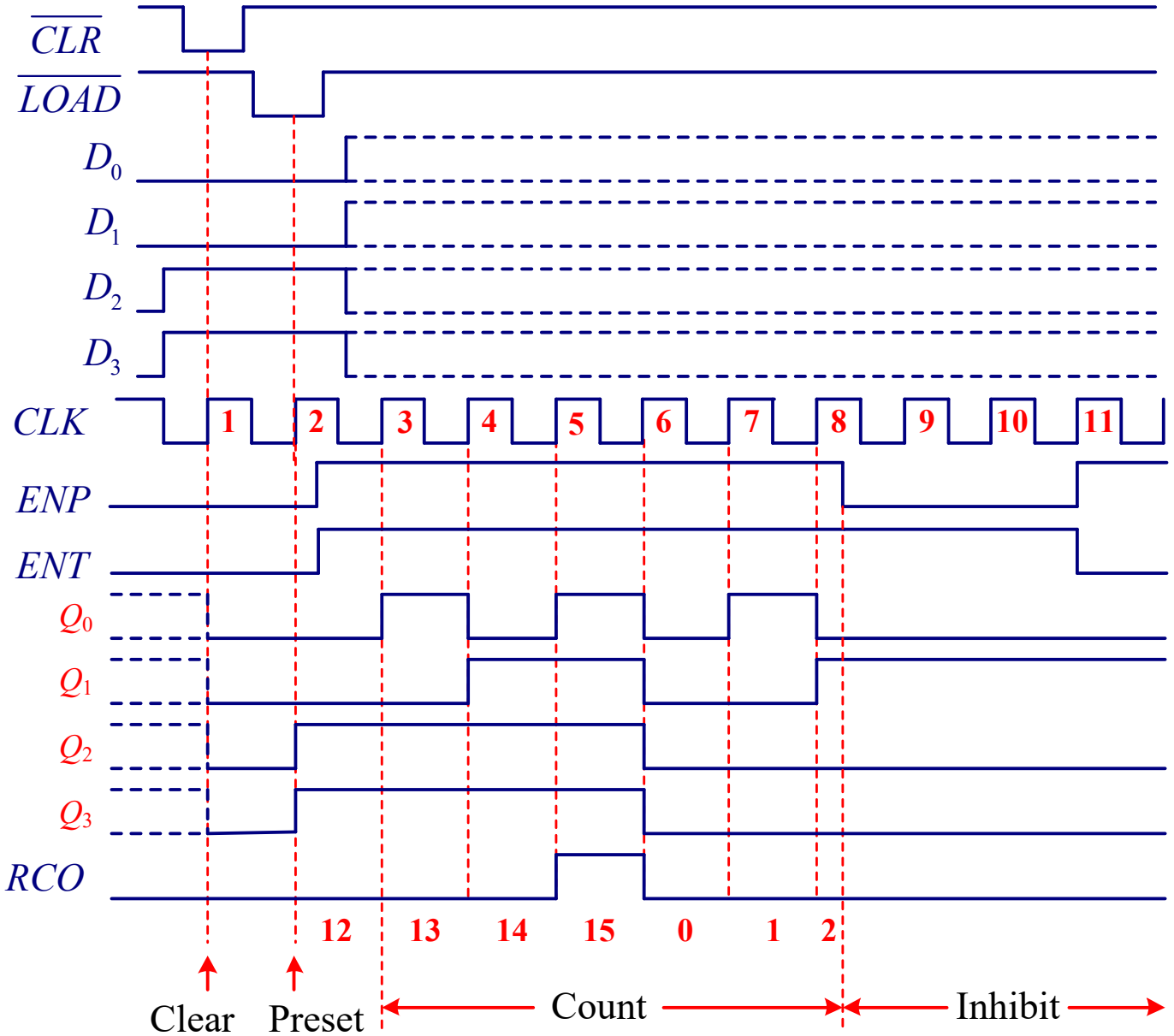
Operating Mode	$\overline{CLR}$	CLK	ENP	ENT	$\overline{LOAD}$	D	Q	RCO
Reset (Clear)	0	↑	X	X	X	X	0	0
Parallel load	1	↑	X	X	0	D	D	0
Count	1	↑	1	1	1	X	Count	†
Inhibit	1	X	0	X	1	X	Q	†
	1	X	X	0	1	X	Q	0

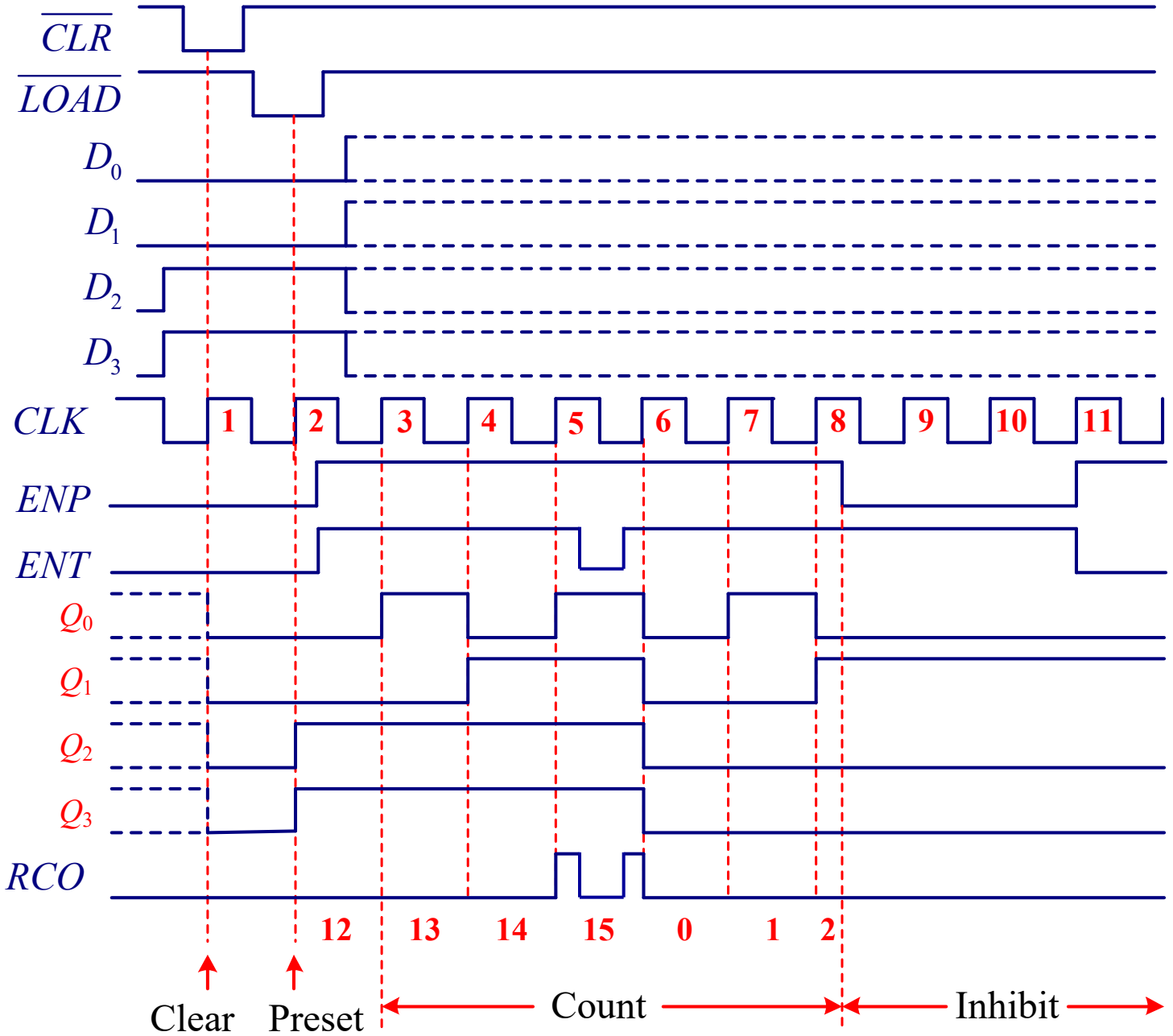
† The RCO output is 1 when ENT is 1 and the counter is at TC (1111).

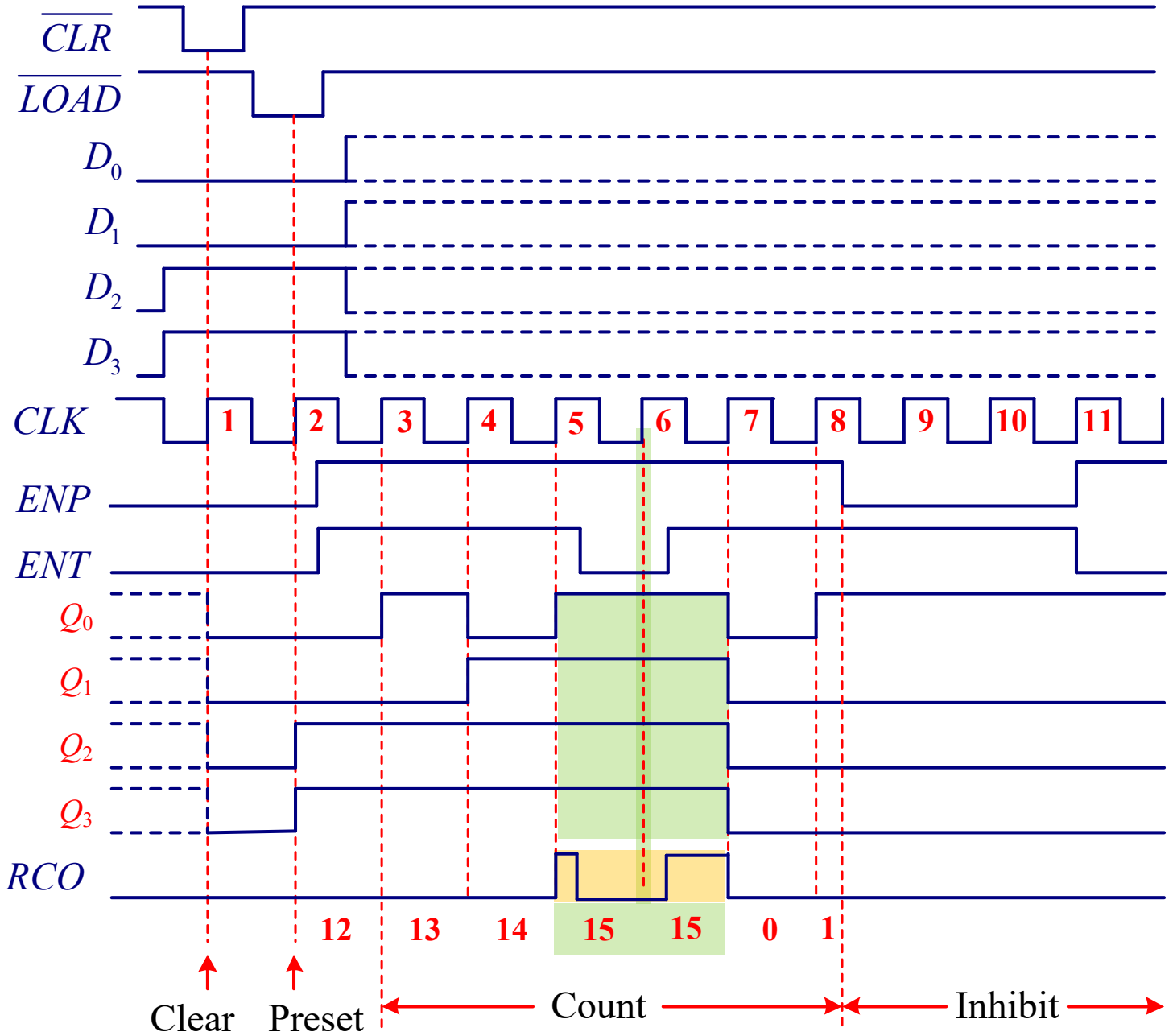
The High-to-Low transition of ENP and ENT should only occur while CLK is HIGH for conventional operation.

The Low-to-High transition of $\overline{CLR}$ and $\overline{LOAD}$ should only occur while CLK is HIGH for conventional operation.

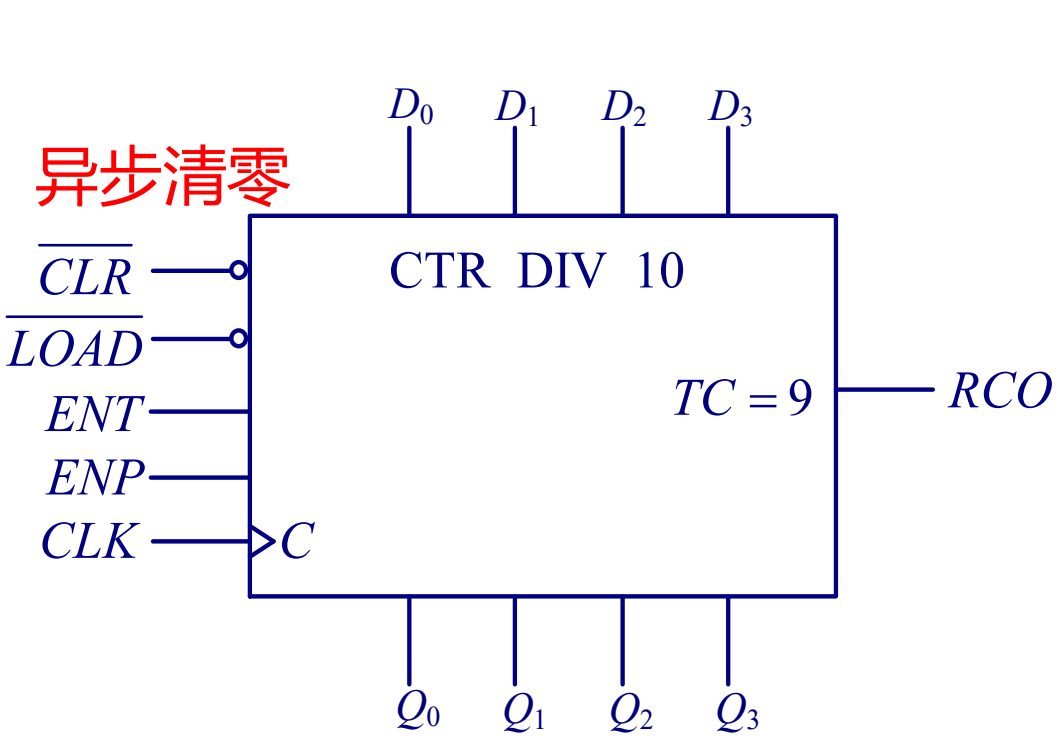
Timing Example for a 74HC163

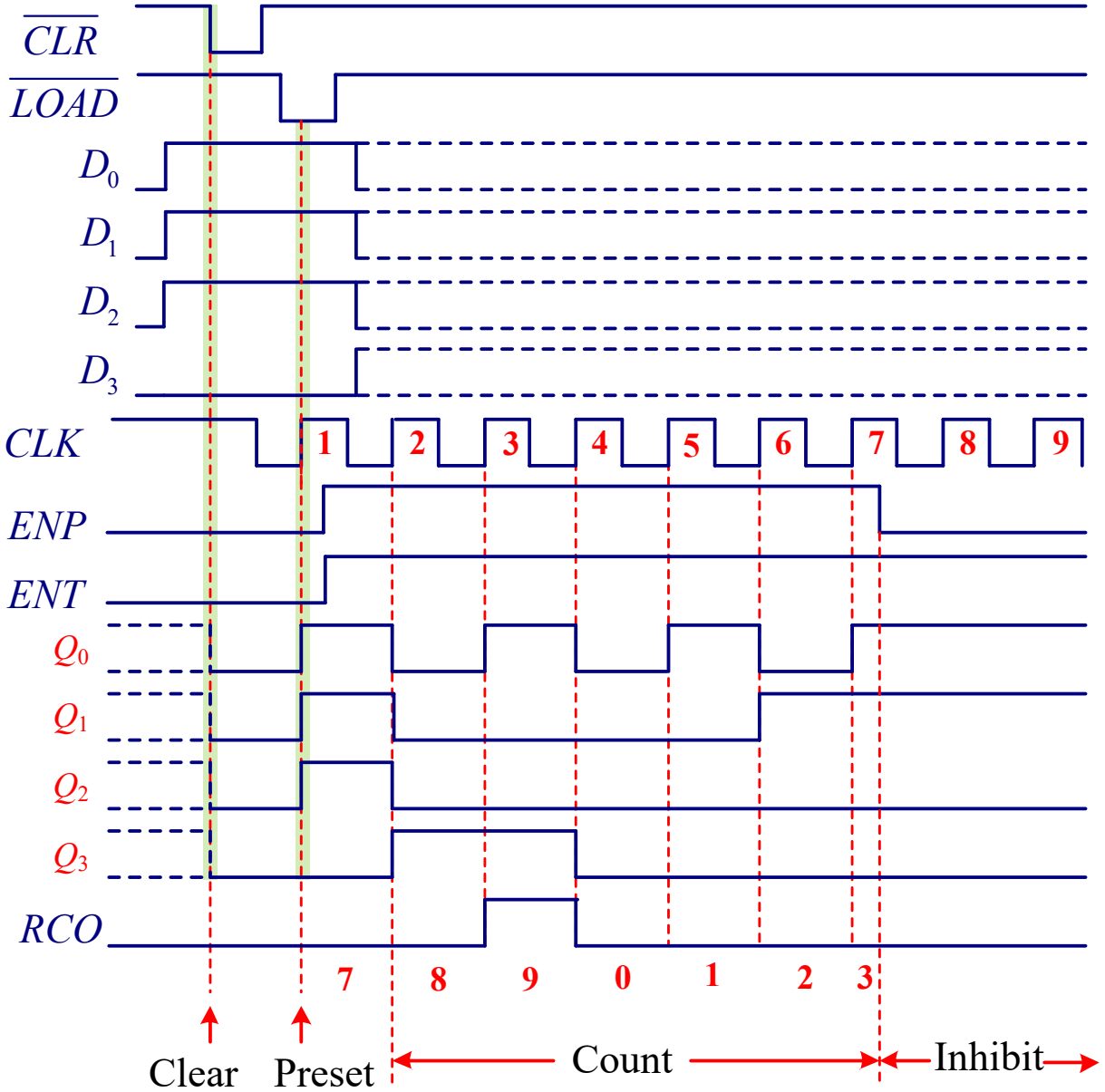



$\overline{CLR}$
$\overline{LOAD}$
D_0
D_1
D_2
D_3
CLK
ENP
ENT
Q_0
Q_1
Q_2
Q_3
RCO
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
0
1
2
Clear
Preset
Count
Inhibit

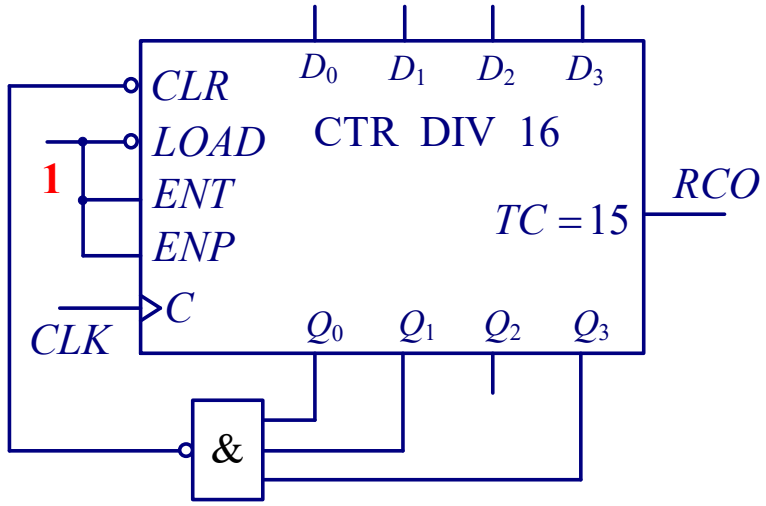

$\overline{CLR}$
$\overline{LOAD}$
D_0
D_1
D_2
D_3
CLK
ENP
ENT
Q_0
Q_1
Q_2
Q_3
RCO
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
15
0
1
Clear
Preset
Count
Inhibit

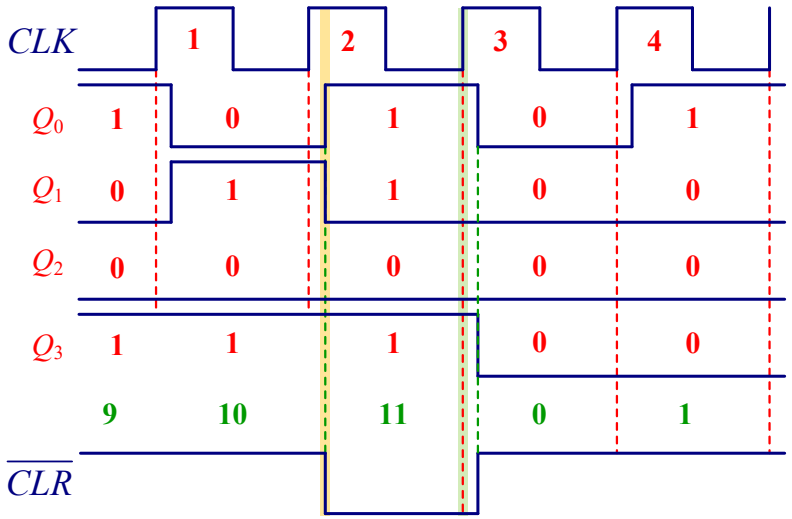
74HC160：Synchronous BCD Decade Counter

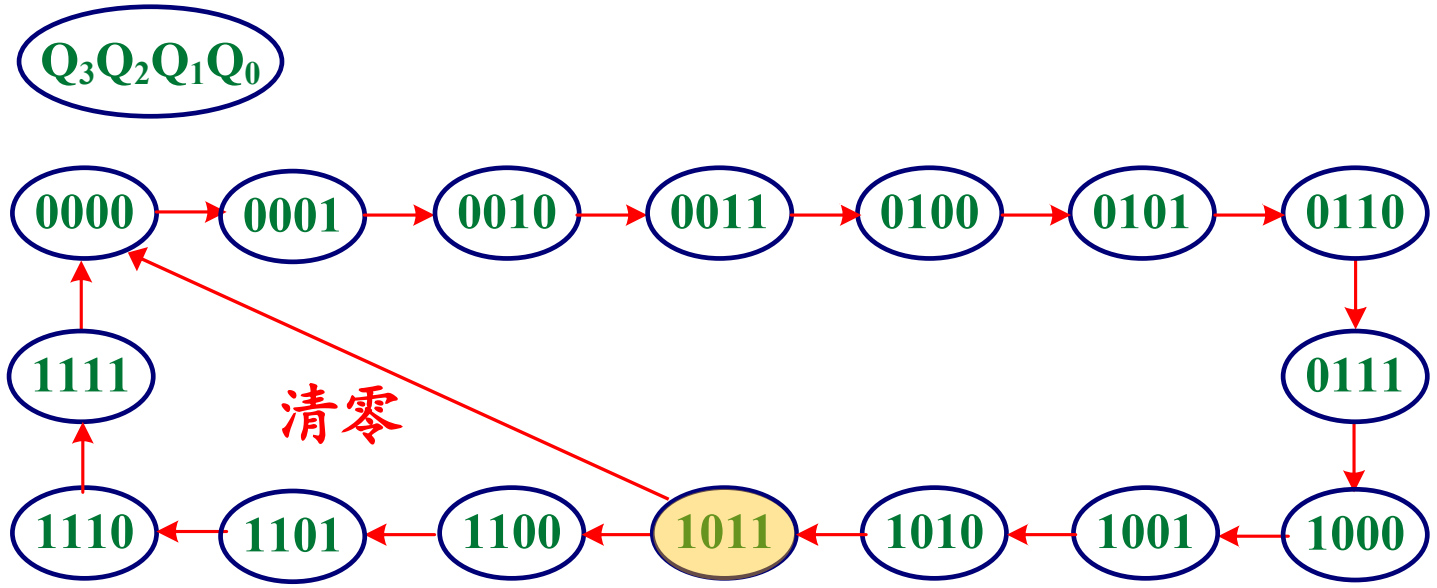




例：用74HC163实现一个模为12的同步计数器

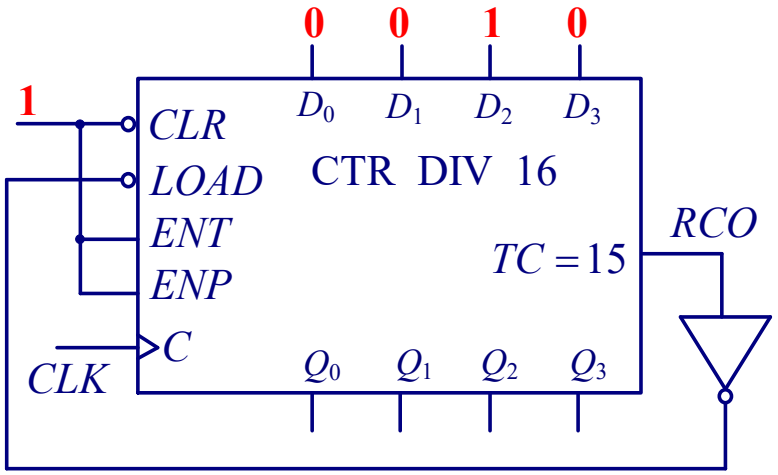


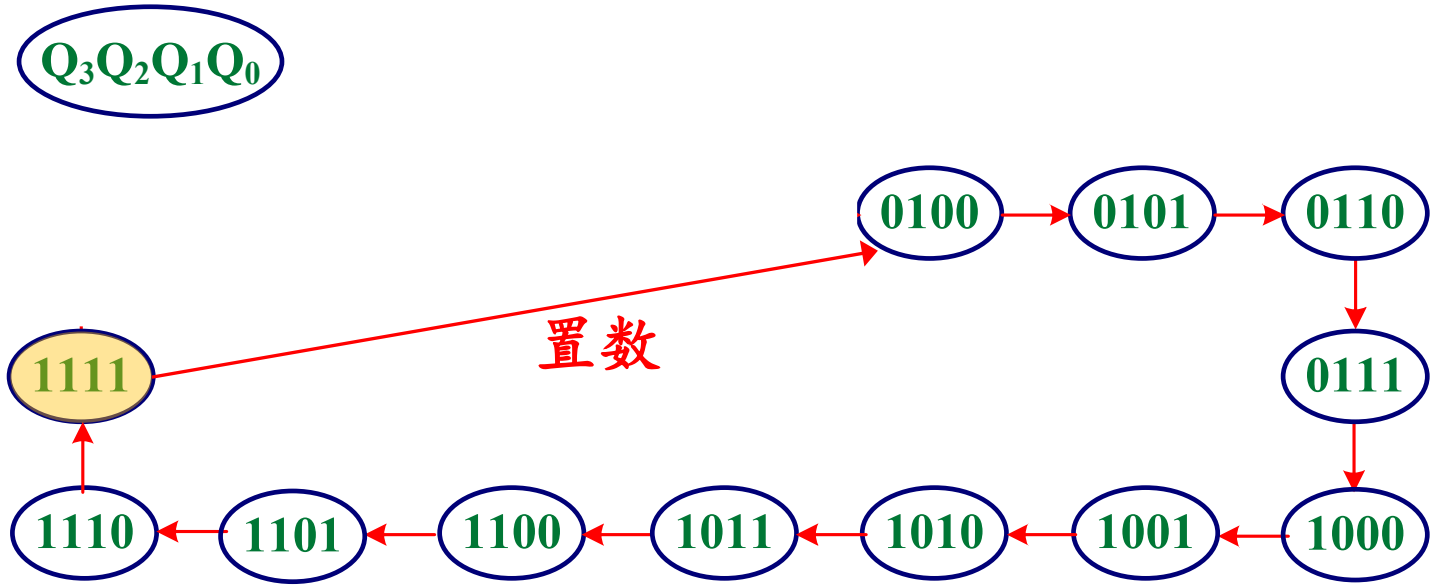




注意：74HC163同步清零，同步置数。

例：用74HC163实现一个模为12的同步计数器

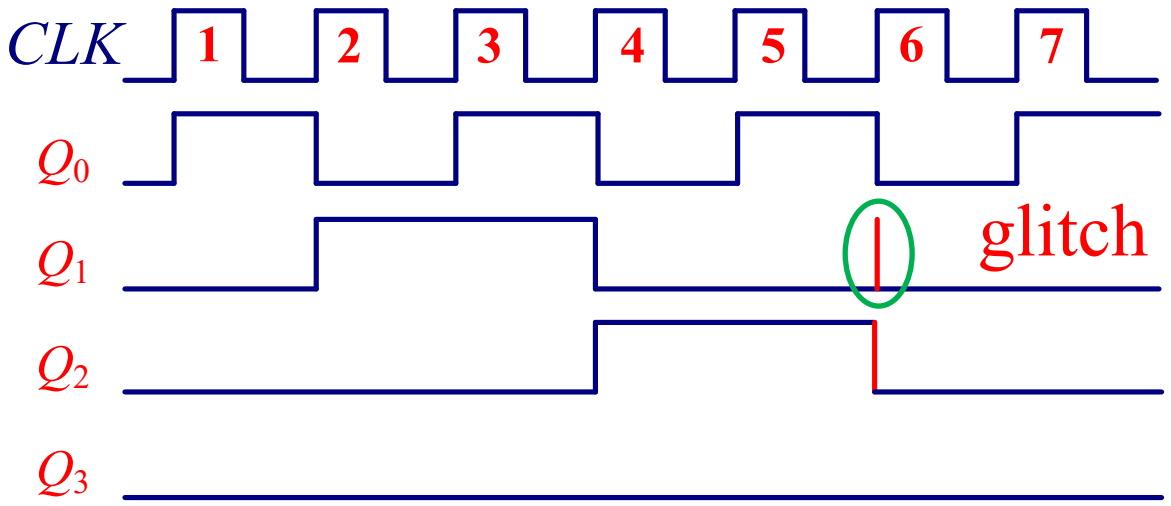


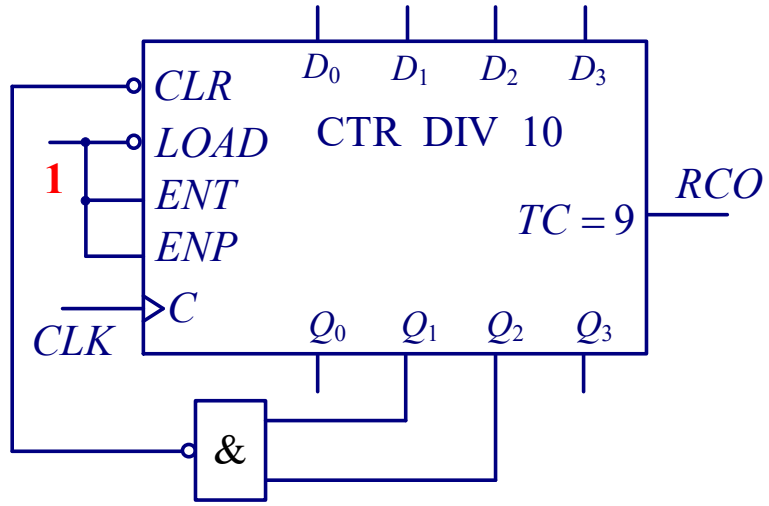


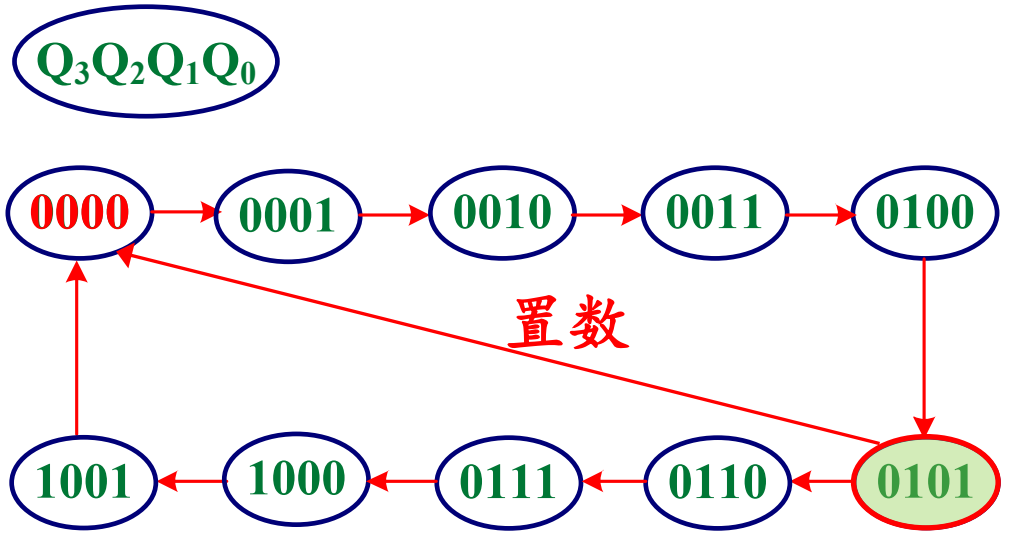
注意：74HC163同步清零，同步置数。

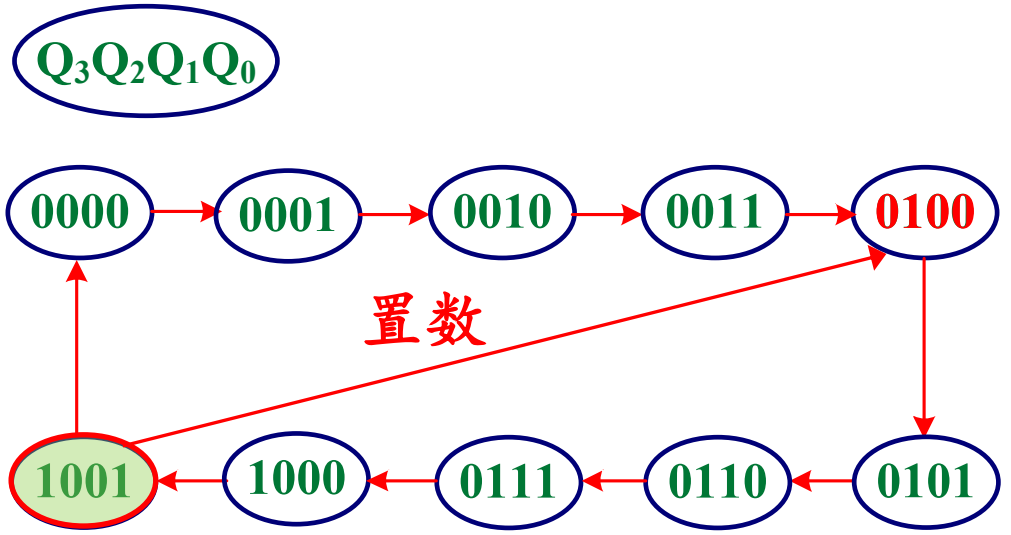
练习：试用74HC160实现模6计数器。

例：用74HC160实现一个模6计数器 （74HC160异步清零，同步置数）

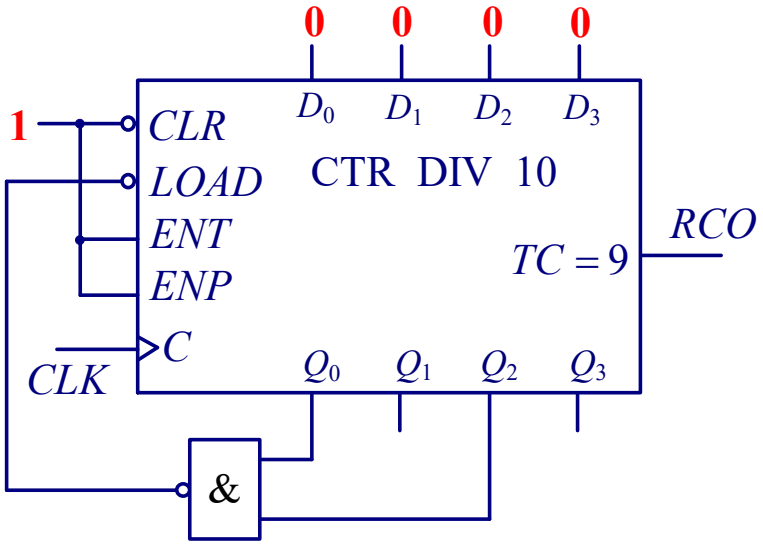


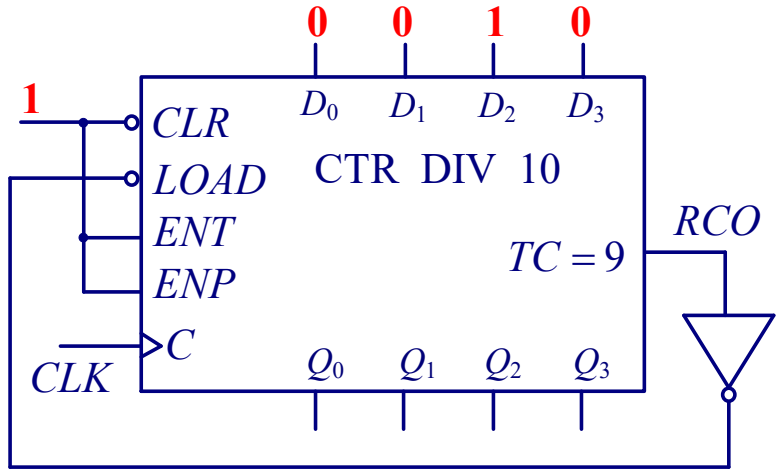


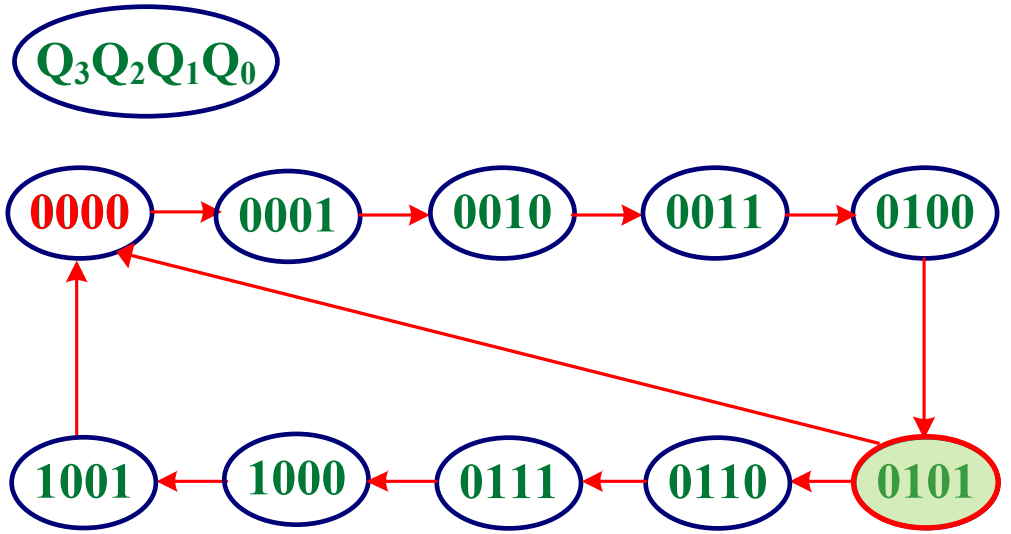


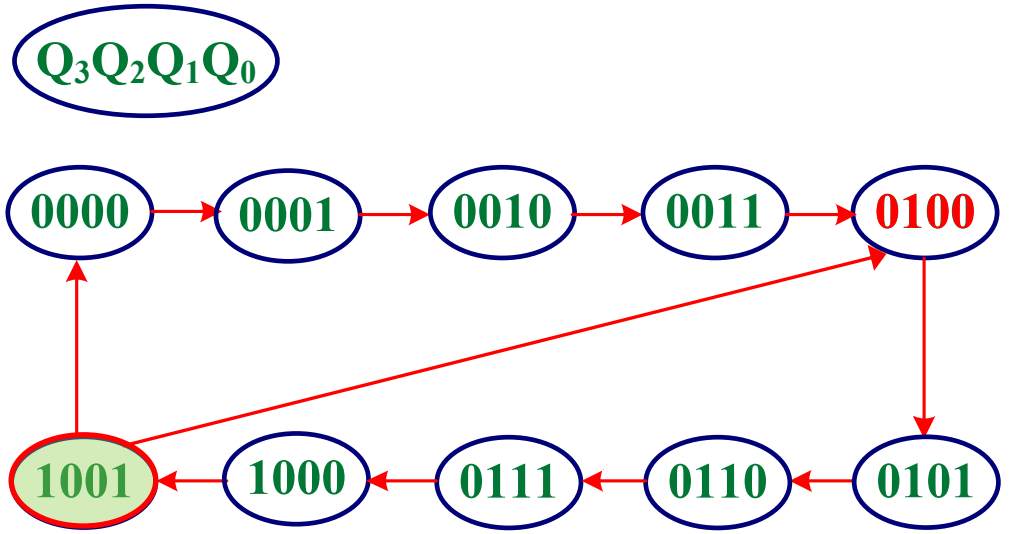


例：用74HC160实现一个模6计数器 （74HC160异步清零，同步置数）

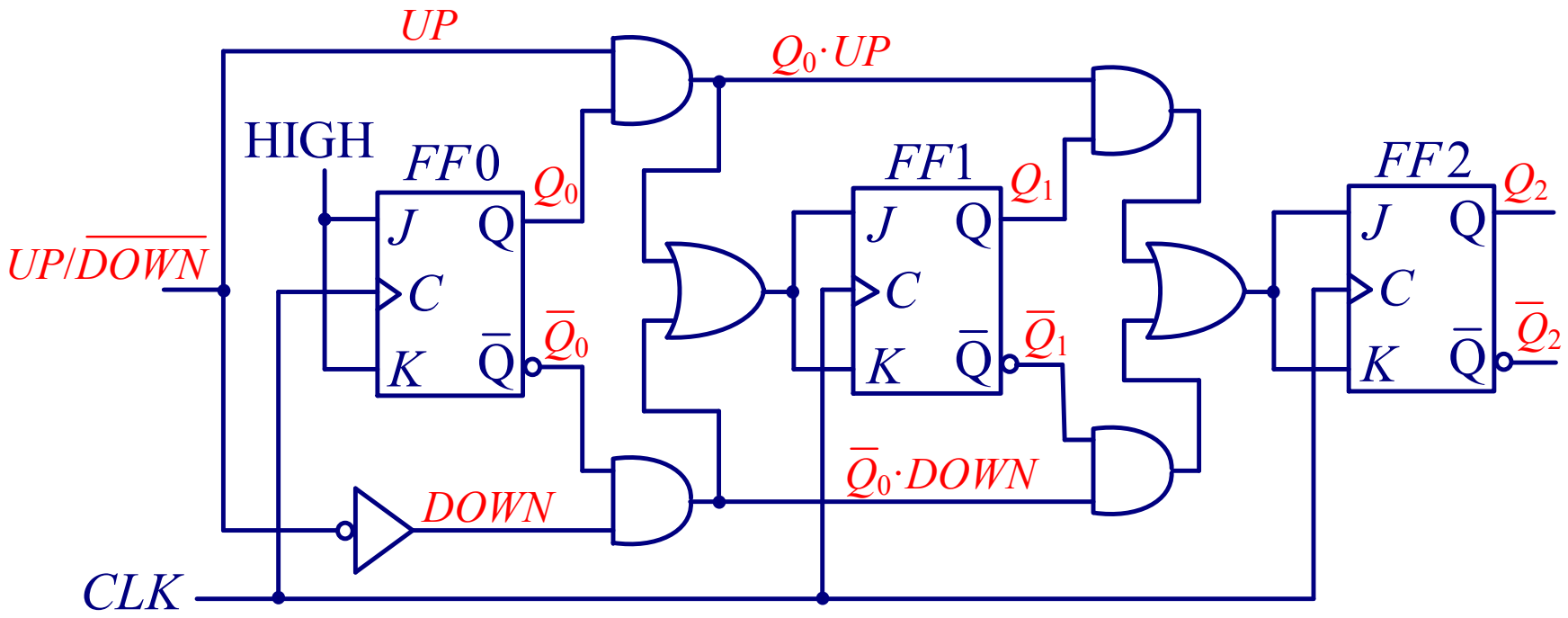








3、双向计数器 Up/Down Counter



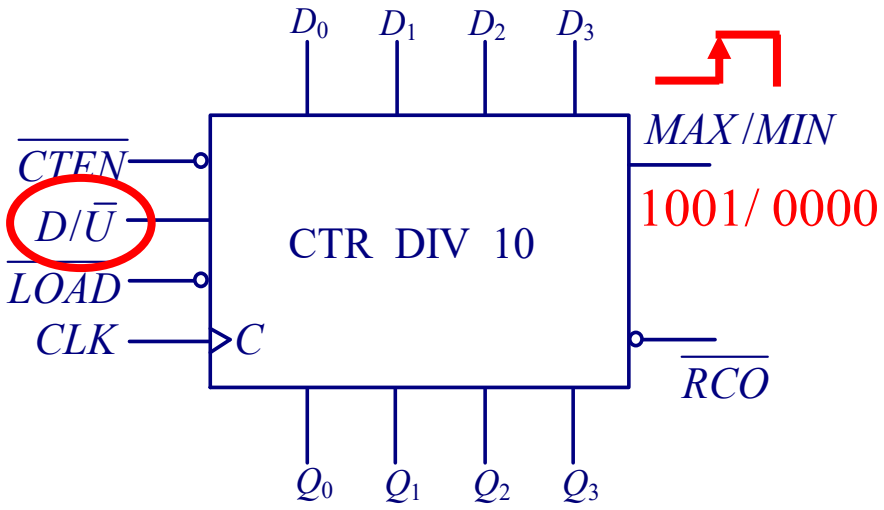
$$J_0 = K_0 = 1$$

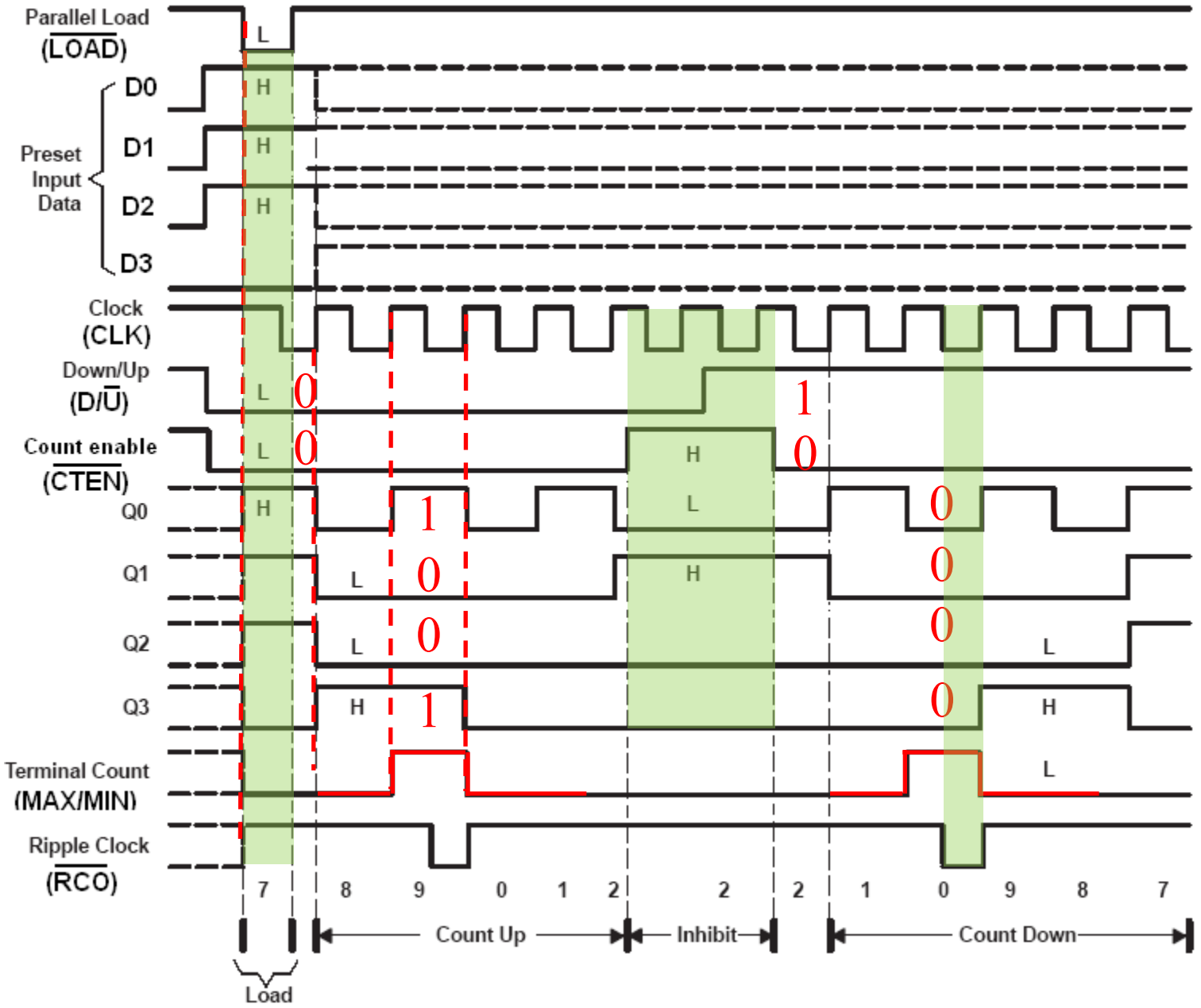
$$J_1 = K_1 = Q_0^n \cdot UP + \overline{Q}_0^n \cdot DOWN$$

$$J_2 = K_2 = Q_0^n \cdot Q_1^n \cdot UP + \overline{Q}_0^n \cdot \overline{Q}_1^n \cdot DOWN$$

74LS190

Synchronous Up/Down Decade Counter





03

时序逻辑电路



3 - 1 锁存器、触发器

3 - 2 计数器

3 - 3 时序逻辑电路分析与设计

一、时序逻辑电路的逻辑功能分析

确定给定电路的驱动方程、输出方程和状态方程

1. 根据逻辑图写出驱动方程和输出方程

时钟方程：各个触发器时钟信号的逻辑表达式；

输出方程：时序电路各个输出信号的逻辑表达式。

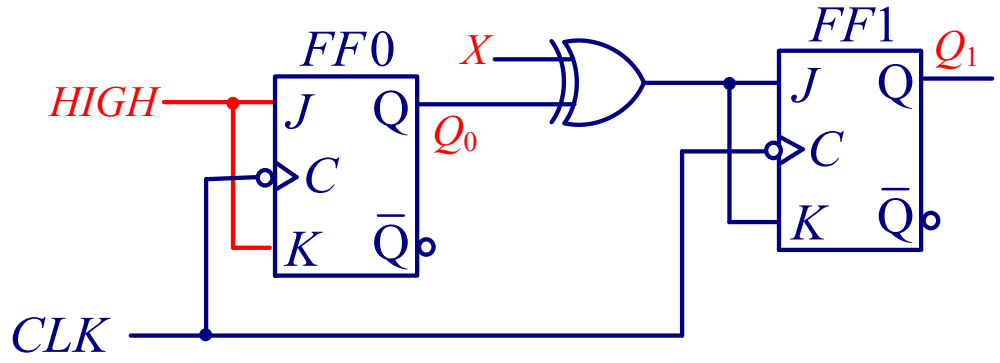
驱动方程：各个触发器输入端信号的逻辑表达式。

2. 求状态方程

状态方程：把驱动方程代入触发器的特性方程

3. 利用状态转换表、状态转换图或时序图等得出结论

例. 分析下图所示电路。画出状态转换图及输出端Q_1、Q_0的波形图。设电路的初始状态为“00”，X为控制端输入信号，可为“0”或“1”。



JK触发器的特性方程为：

$$Q^{n+1} = J\bar{Q}^n + \bar{K}Q^n$$

解：此为同步时序逻辑电路

驱动方程：$J_0 = K_0 = 1$

$$J_1 = K_1 = X \oplus Q_0^n$$

状态方程：

$$Q_0^{n+1} = J_0\bar{Q}_0^n + \bar{K}_0 Q_0^n = \bar{Q}_0^n$$

$$Q_1^{n+1} = \left(X \oplus Q_0^n\right)\bar{Q}_1^n + \overline{X \oplus Q_0^n}\,Q_1^n$$

$$= X \oplus Q_0^n \oplus Q_1^n$$

(1) 触发器的状态方程

$$Q_0^{n+1} = \bar{Q}_0^n$$

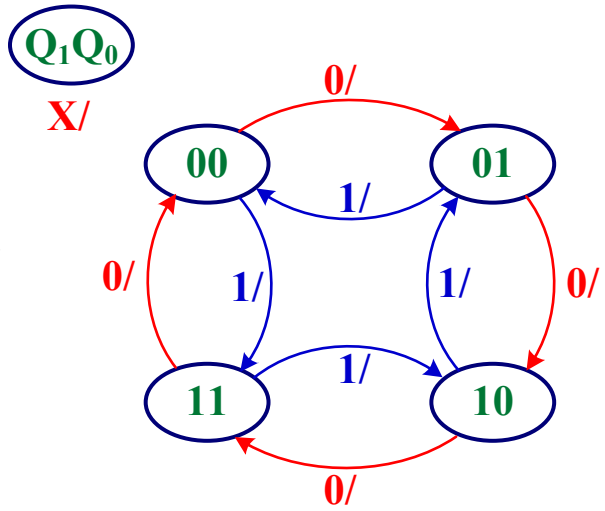
$$Q_1^{n+1} = X \oplus Q_0^n \oplus Q_1^n$$

(2) 由触发器的状态方程列出状态转换表

(3) 画出状态转换图

X=0，同步四进制加法计数器

X=1，同步四进制减法计数器



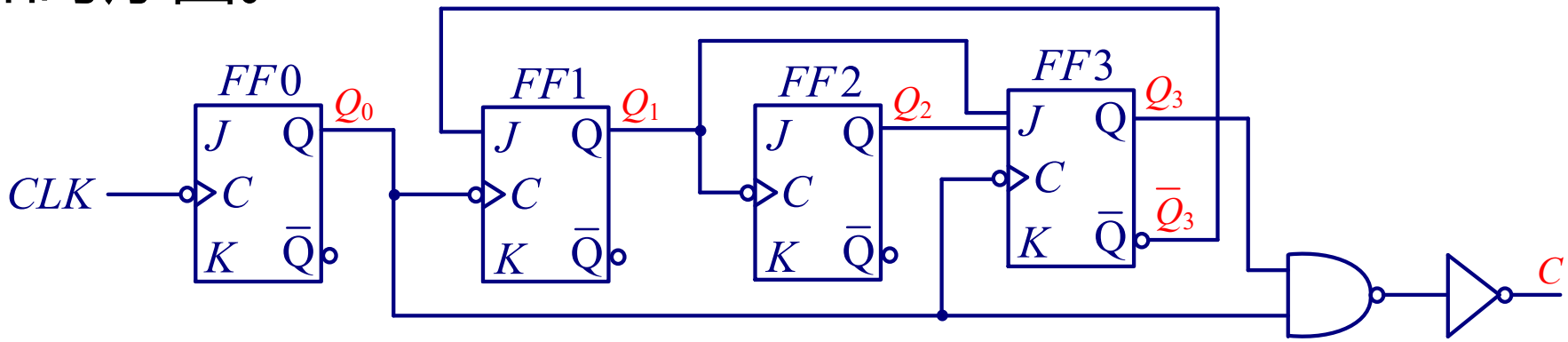
state diagram

Next-state table

状态转换表

X	Q_1	Q_0
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	1	1
1	1	0
1	0	1

例：已知电路逻辑图如下，试分析其逻辑功能，并画出电路的状态转换图和时序图。



解：此为异步时序逻辑电路

驱动方程：

$$\begin{cases} J_0 = K_0 = 1 \\ J_1 = \bar{Q}_3^n, \quad K_1 = 1 \\ J_2 = K_2 = 1 \\ J_3 = Q_1^n Q_2^n, \quad K_3 = 1 \end{cases}$$

JK触发器的特性方程为：$Q^{n+1} = J\bar{Q}^n + \bar{K}Q^n$

每个触发器的状态方程：

$$\begin{cases} Q_0^{n+1} = \bar{Q}_0^n & (CLK\downarrow) \\ Q_1^{n+1} = \bar{Q}_3^n \bar{Q}_1^n & (Q_0\downarrow) \\ Q_2^{n+1} = \bar{Q}_2^n & (Q_1\downarrow) \\ Q_3^{n+1} = Q_1^n Q_2^n \bar{Q}_3^n & (Q_0\downarrow) \end{cases}$$

输出方程：$C = Q_3^n Q_0^n$

$$\begin{cases} Q_0^{n+1} = \bar{Q}_0^n & (CLK\downarrow) \\ Q_1^{n+1} = \bar{Q}_3^n \bar{Q}_1^n & (Q_0\downarrow) \\ Q_2^{n+1} = \bar{Q}_2^n & (Q_1\downarrow) \\ Q_3^{n+1} = Q_1^n Q_2^n \bar{Q}_3^n & (Q_0\downarrow) \end{cases}$$

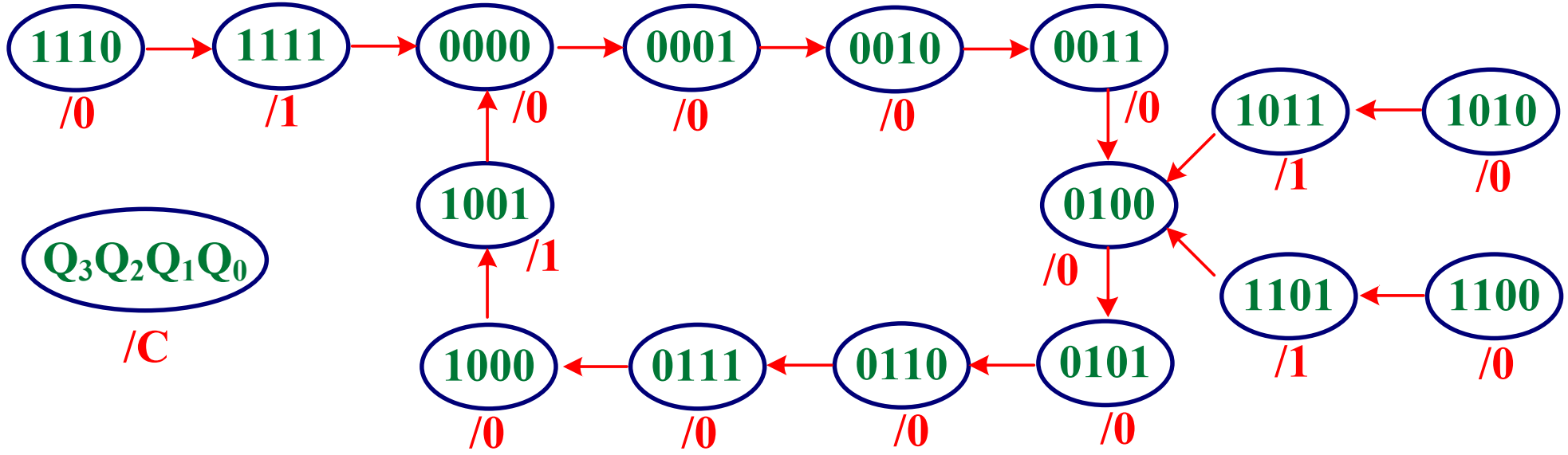
$$C = Q_3^n Q_0^n$$

电路的状态转换表

CLK	Q_3^n	Q_2^n	Q_1^n	Q_0^n	C
0	0	0	0	0	0
1	0	0	0	1	0
2	0	0	1	0	0
3	0	0	1	1	0
4	0	1	0	0	0
5	0	1	0	1	0
6	0	1	1	0	0
7	0	1	1	1	0
8	1	0	0	0	0
9	1	0	0	1	1

CLK	Q_3^n	Q_2^n	Q_1^n	Q_0^n	C
0	1	0	1	0	0
1	1	0	1	1	1
2	0	1	0	0	0
0	1	1	0	0	0
1	1	1	0	1	1
2	0	1	0	0	0
0	1	1	1	0	0
1	1	1	1	1	1
2	0	0	0	0	0

画出电路状态转换图如下



电路逻辑功能：可以自行启动的异步十进制加法计数器

二、任意进制计数器的设计

计数器的设计方法很多，大致可分为两类：一是根据要求用触发器构成，再就是利用具有特定功能的中规模集成器件适当连接而成。

A. 利用触发器设计计数电路

例：数字控制装置中常用的步进电动机有 A、B、C 三个绕组。电动机运行时要求三个绕组以 A→AB →B →BC →C →CA→A的顺序循环通电，试设计一个电路实现之。

例：数字控制装置中常用的步进电动机有 A、B、C 三个绕组。电动机运行时要求三个绕组以 A→AB →B →BC →C →CA→A的顺序循环通电，试设计一个电路实现之。

解：(1) 根据要求确定计数器的模数和所需的触发器个数。

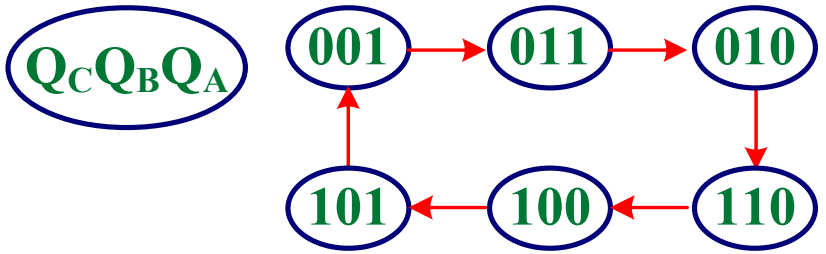
本任务所需计数器模数为 6，所以触发器的个数为 3 。

(2) 确定触发器的类型。

最常用的触发器有 D触发器和JK触发器，本任务中选用JK触发器。

(3) 列写状态转换表或状态图。

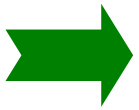
用三个触发器的输出端Q_A、Q_B、Q_C分别控制电动机的三个绕组A、B、C，并以“1”表示通电，“0”表示不通电。



(4) 根据所选触发器的功能表，确定各个触发器在状态转换时对控制端的电平要求（触发器转换表）。

JK触发器的功能表

J	K	Q^n	Q^{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0



JK触发器的转换表

$Q^n \rightarrow Q^{n+1}$	J	K
0 0	0	X
0 1	1	X
1 0	X	1
1 1	X	0

注：“X”表示可“0”可“1”

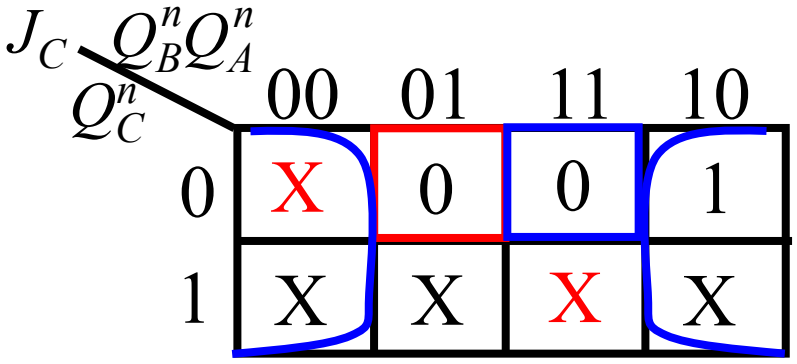
JK触发器的转换表

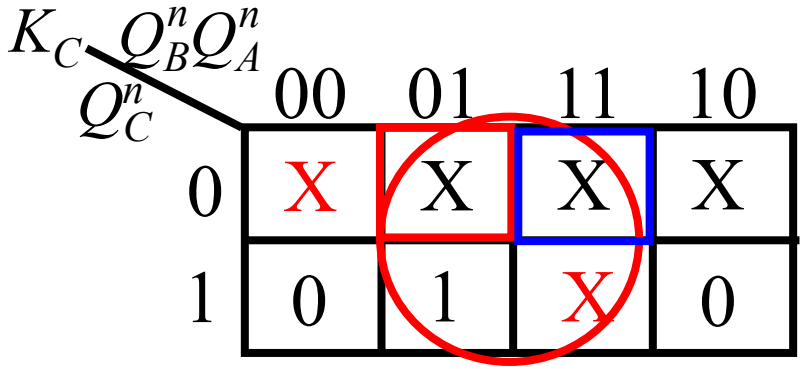
$Q^n \rightarrow Q^{n+1}$		J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

(5) 根据题意列出次态表及对应的触发器输入逻辑

原状态			下状态			对各控制端的电平要求					
Q_C^n	Q_B^n	Q_A^n	Q_C^{n+1}	Q_B^{n+1}	Q_A^{n+1}	J_C	K_C	J_B	K_B	J_A	K_A
0	0	1	0	1	1	0	X	1	X	X	0
0	1	1	0	1	0	0	X	X	0	X	1
0	1	0	1	1	0	1	X	X	0	0	X
1	1	0	1	0	0	X	0	X	1	0	X
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	0	0	1	X	1	0	X	X	0

(6) 填写卡诺图，以得出各个触发器输入逻辑表达式。





$$J_C = \bar{Q}_A^n, \quad K_C = Q_A^n$$

原 状 态			下 状 态			对各控制端的电平要求					
Q_C^n	Q_B^n	Q_A^n	Q_C^{n+1}	Q_B^{n+1}	Q_A^{n+1}	J_C	K_C	J_B	K_B	J_A	K_A
0	0	1	0	1	1	0	X	1	X	X	0
0	1	1	0	1	0	0	X	X	0	X	1
0	1	0	1	1	0	1	X	X	0	0	X
1	1	0	1	0	0	X	0	X	1	0	X
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	0	0	1	X	1	0	X	X	0

J_B / Q_C^n \ $Q_B^n Q_A^n$

Q_C^n	00	01	11	10
0	X	1	X	X
1	0	0	X	X

K_B / Q_C^n \ $Q_B^n Q_A^n$

Q_C^n	00	01	11	10
0	X	X	0	0
1	X	X	X	1

J_A / Q_C^n \ $Q_B^n Q_A^n$

Q_C^n	00	01	11	10
0	X	X	X	0
1	1	X	X	0

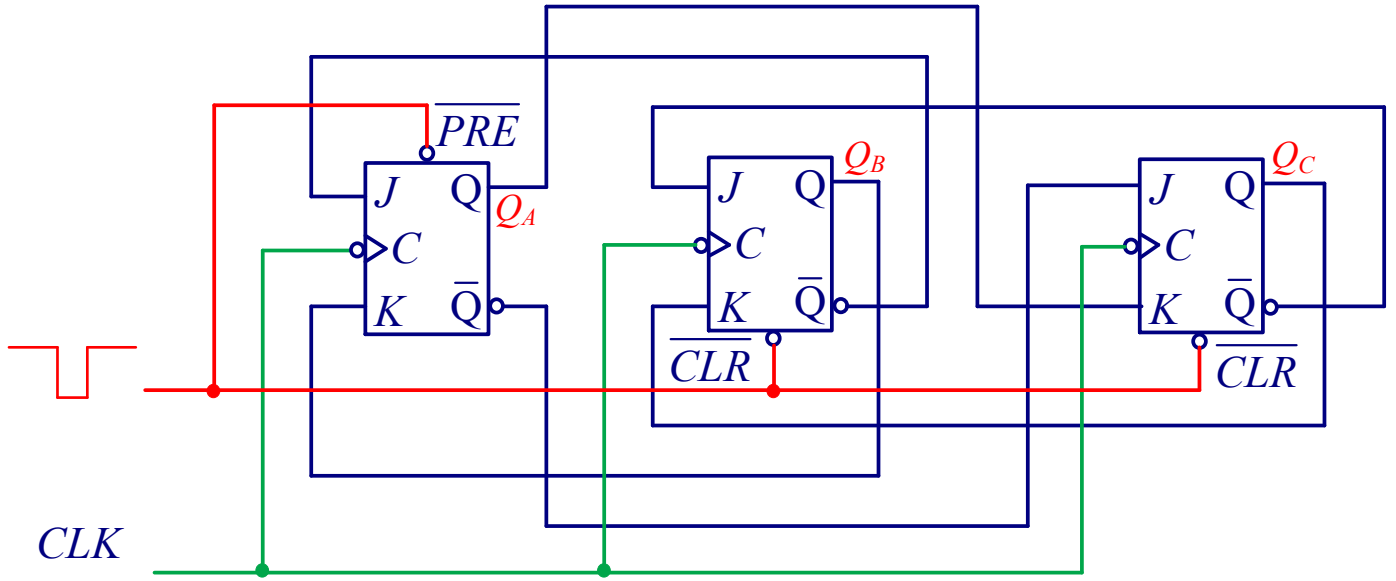
K_A / Q_C^n \ $Q_B^n Q_A^n$

Q_C^n	00	01	11	10
0	X	0	1	X
1	X	0	X	X

$$J_B = \bar{Q}_C^n, \quad K_B = Q_C^n$$

$$J_A = \bar{Q}_B^n, \quad K_A = Q_B^n$$

(7) 画出计数器的逻辑电路图。



$$J_A = \bar{Q}_B^n, \quad K_A = Q_B^n$$

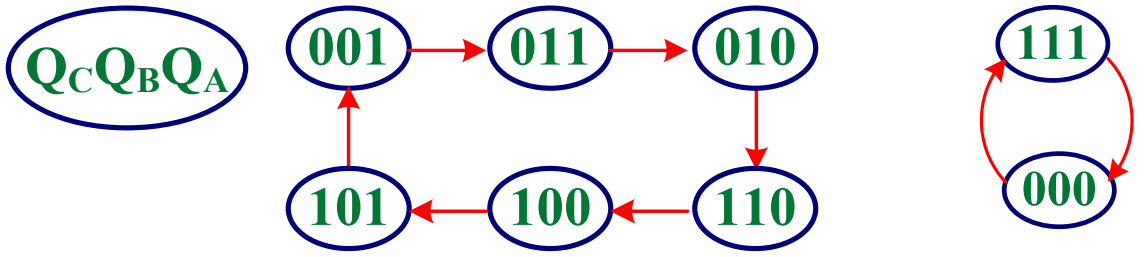
$$J_B = \bar{Q}_C^n, \quad K_B = Q_C^n$$

$$J_C = \bar{Q}_A^n, \quad K_C = Q_A^n$$

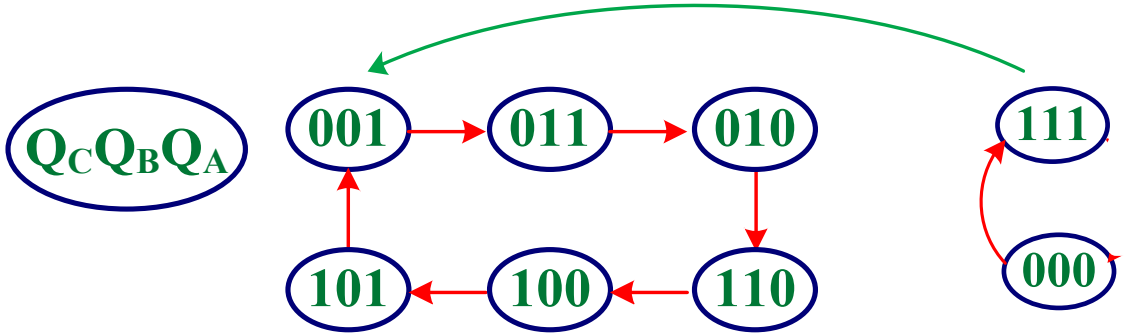
(8) 检验该计数电路能否自启动

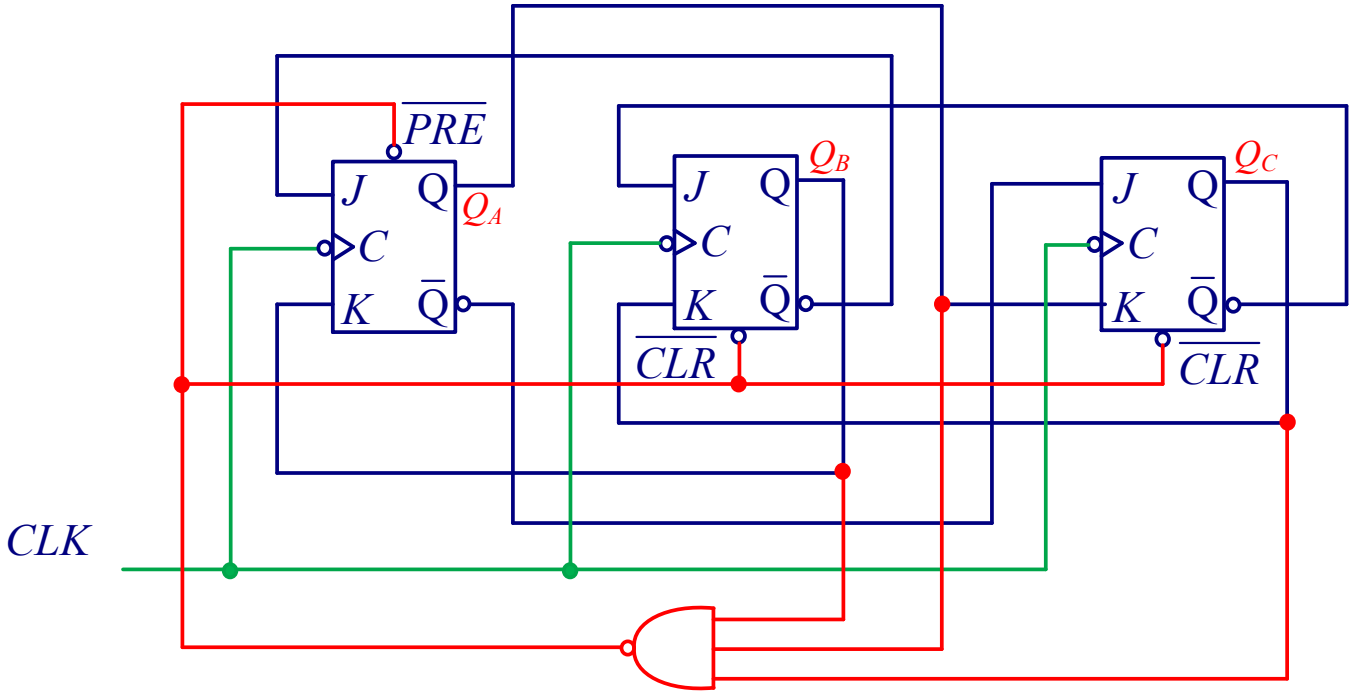
(9) 修改电路

方法一：



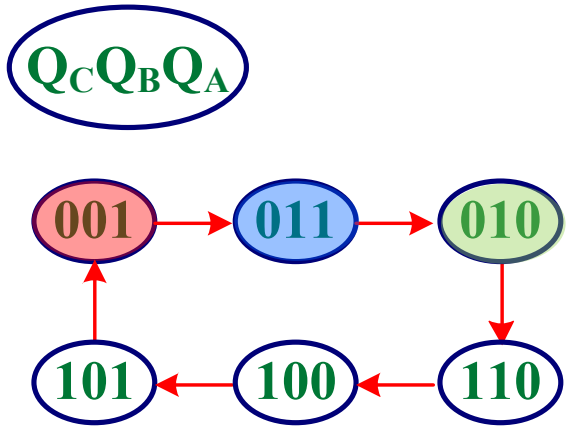
(9) 修改电路


$Q_CQ_BQ_A$
001
011
010
111
101
100
110
000


$\overline{PRE}$
J
Q
C
K
$\overline{Q}$
Q_A
Q_B
$\overline{CLR}$
Q_C
CLK

方法一：

第二种设计过程



这样设计的结果与第一种方法相同，也不能自启动

Q_C^{n+1} / Q_C^n	00	01	11	10
0	X	0	0	1
1	1	0	X	1

$$Q_C^{n+1} = J_C\bar{Q}_C^n + \bar{K}_C Q_C^n$$

$$Q_C^{n+1} = \bar{Q}_A^n$$

$$= \bar{Q}_A^n(\bar{Q}_C^n + Q_C^n)$$

$$= \bar{Q}_A^n\bar{Q}_C^n + \bar{Q}_A^n Q_C^n$$

Q_B^{n+1} / Q_C^n	00	01	11	10
0	X	1	1	1
1	0	0	X	0

$$\therefore J_C = \bar{Q}_A^n,\ K_C = Q_A^n$$

同理：

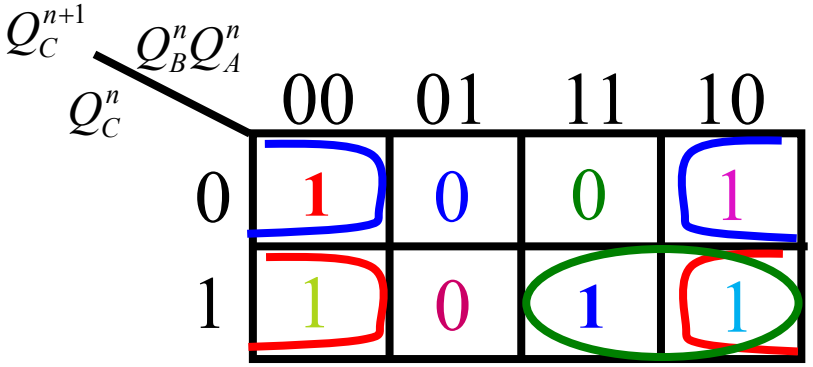
$$Q_B^{n+1} = \bar{Q}_C^n\bar{Q}_B^n + \bar{Q}_C^n Q_B^n$$

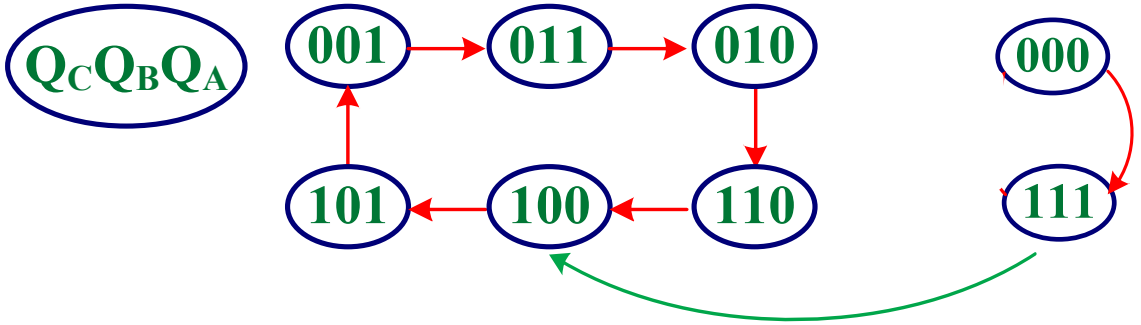
$$Q_A^{n+1} = \bar{Q}_B^n\bar{Q}_A^n + \bar{Q}_B^n Q_A^n$$

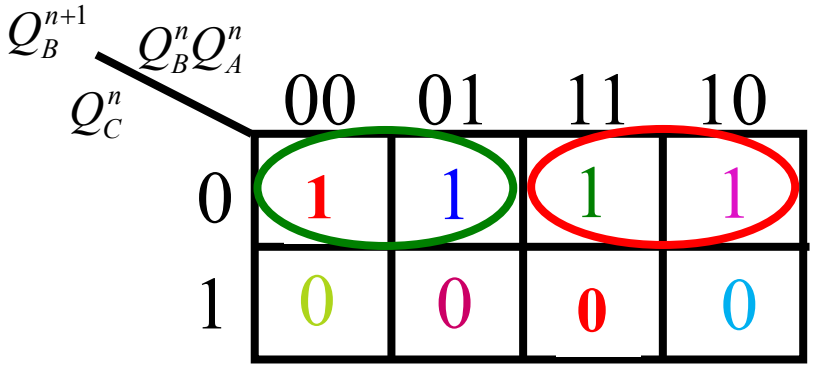
Q_A^{n+1} / Q_C^n	00	01	11	10
0	X	1	0	0
1	1	1	X	0

$$J_B = \bar{Q}_C^n,\ K_B = Q_C^n$$

$$J_A = \bar{Q}_B^n,\ K_A = Q_B^n$$





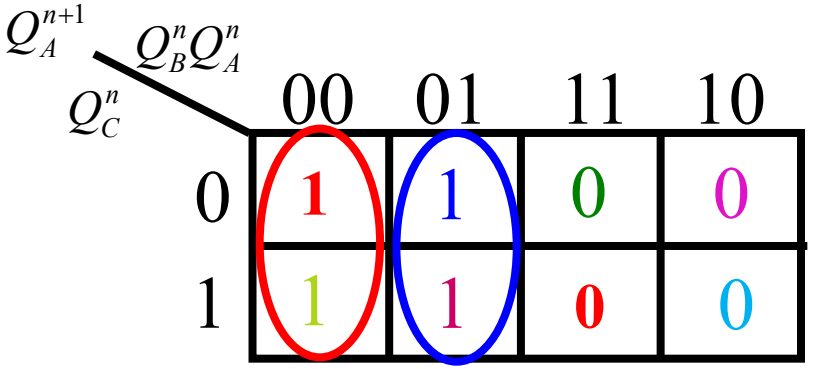


$$Q_C^{n+1} = Q_C^nQ_B^n + Q_C^n\bar{Q}_A^n + \bar{Q}_C^n\bar{Q}_A^n$$
$$= \bar{Q}_A^n\bar{Q}_C^n + (\bar{Q}_A^n + Q_B^n)Q_C^n$$

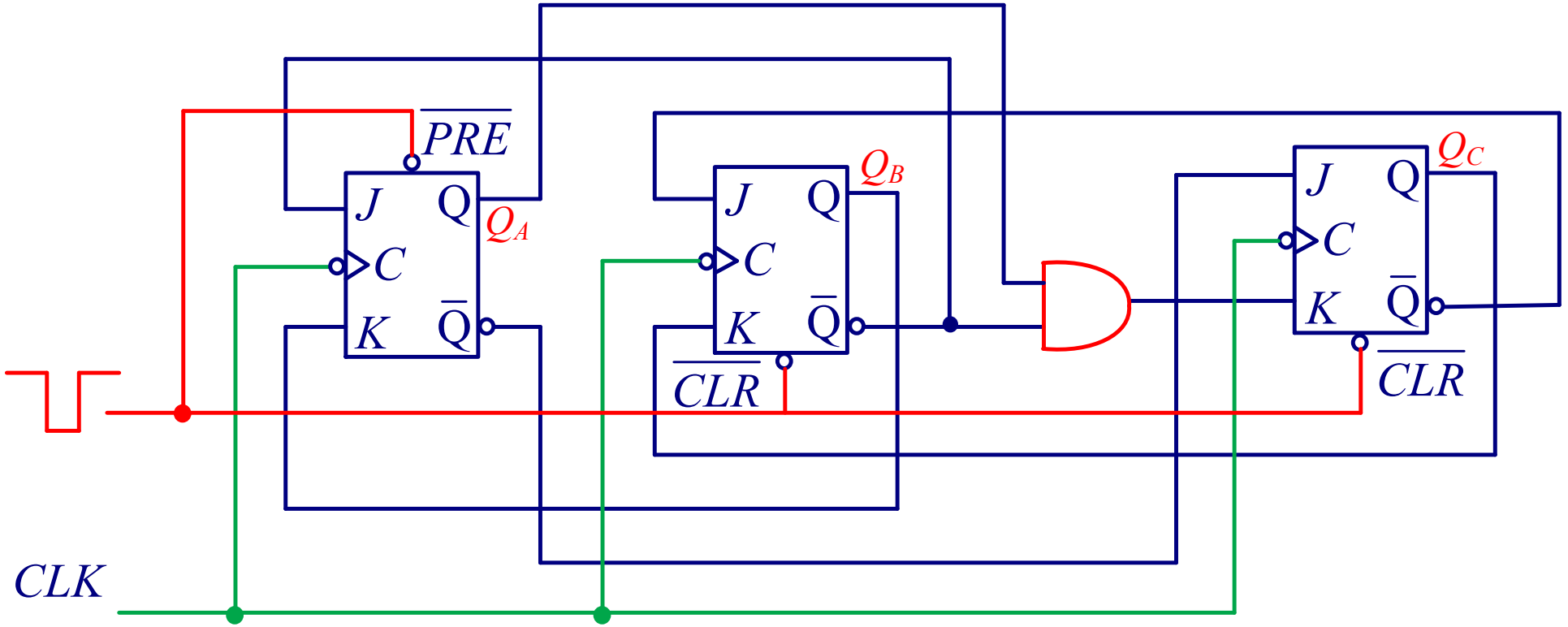
$$Q_C^{n+1} = J_C\bar{Q}_C^n + \bar{K}_CQ_C^n$$

$$\therefore J_C = \bar{Q}_A^n$$

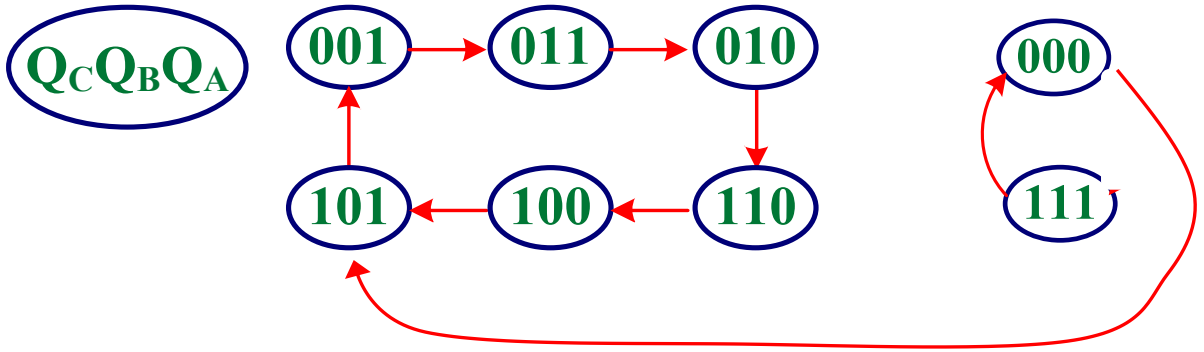
$$K_C = \overline{\bar{Q}_A^n + Q_B^n} = Q_A^n\bar{Q}_B^n$$



$$J_A = \bar{Q}_B, \quad K_A = Q_B \qquad J_B = \bar{Q}_C, \quad K_B = Q_C \qquad J_C = \bar{Q}_A^n, \quad K_C = Q_A^n \bar{Q}_B^n$$



Q_C^{n+1} / $Q_B^nQ_A^n$ / Q_C^n	00	01	11	10
0	X	0	0	1
1	1	0	X	1



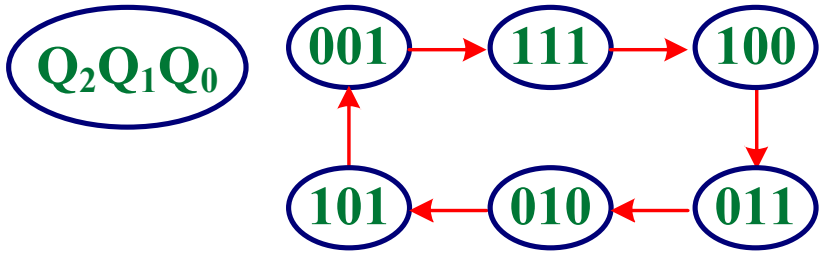
Q_B^{n+1} / $Q_B^nQ_A^n$ / Q_C^n	00	01	11	10
0	0	1	1	1
1	0	0	X	0

$$Q_B^{n+1} = \bar{Q}_C^n Q_A^n \bar{Q}_B^n + \bar{Q}_C^n Q_B^n$$

$$J_B = \bar{Q}_C^n Q_A^n, \quad K_B = Q_C^n$$

Q_A^{n+1} / $Q_B^nQ_A^n$ / Q_C^n	00	01	11	10
0	X	1	0	0
1	1	1	X	0

Design a synchronous counter with the binary sequence 001,111, 100, 011, 010, 101, 001…. Use JK flip-flops and gates.



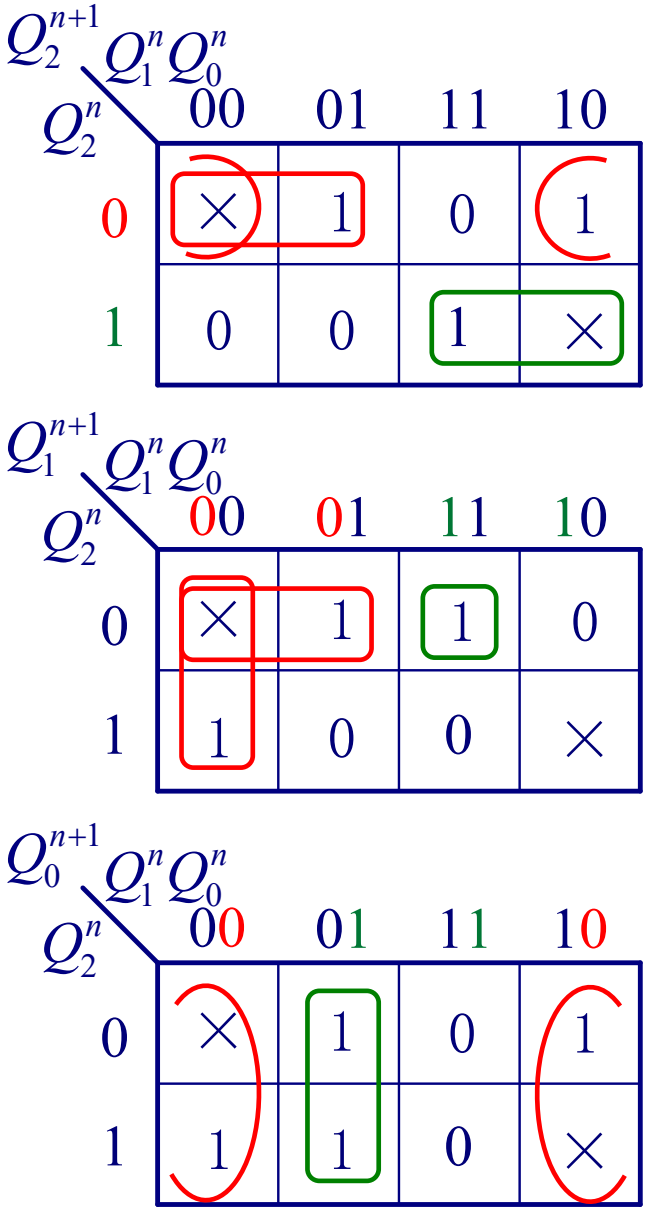
$$Q^{n+1} = J\bar{Q}^n + \bar{K}Q^n$$

$$Q_2^{n+1} = (\bar{Q}_0^n + \bar{Q}_1^n)\bar{Q}_2^n + Q_1^n Q_2^n$$

$$J_2 = \bar{Q}_0^n + \bar{Q}_1^n = \overline{Q_0^n Q_1^n} \qquad K_2 = \bar{Q}_1^n$$

$$Q_1^{n+1} = (\bar{Q}_0^n + \bar{Q}_2^n)\bar{Q}_1^n + \bar{Q}_2^n Q_0^n Q_1^n$$

$$Q_0^{n+1} = \bar{Q}_0^n + \bar{Q}_1^n Q_0^n$$

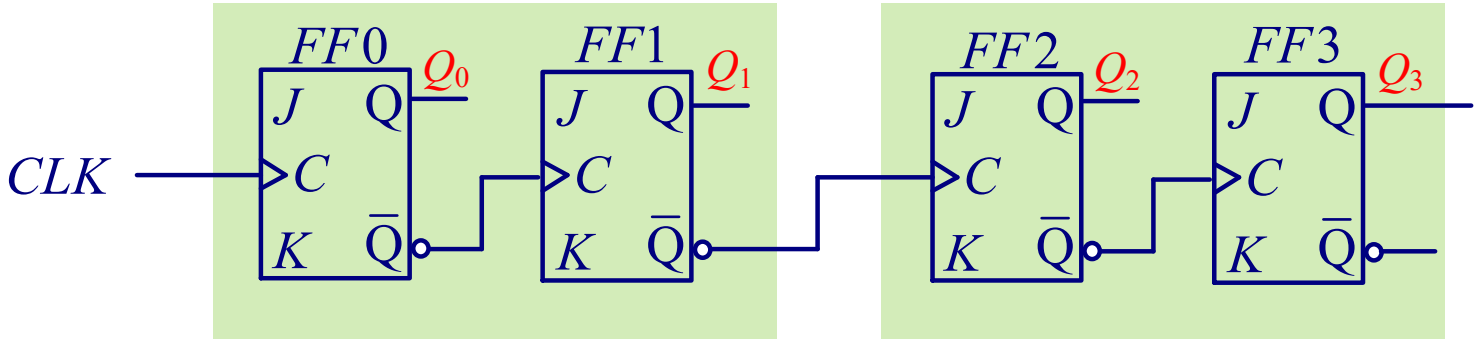


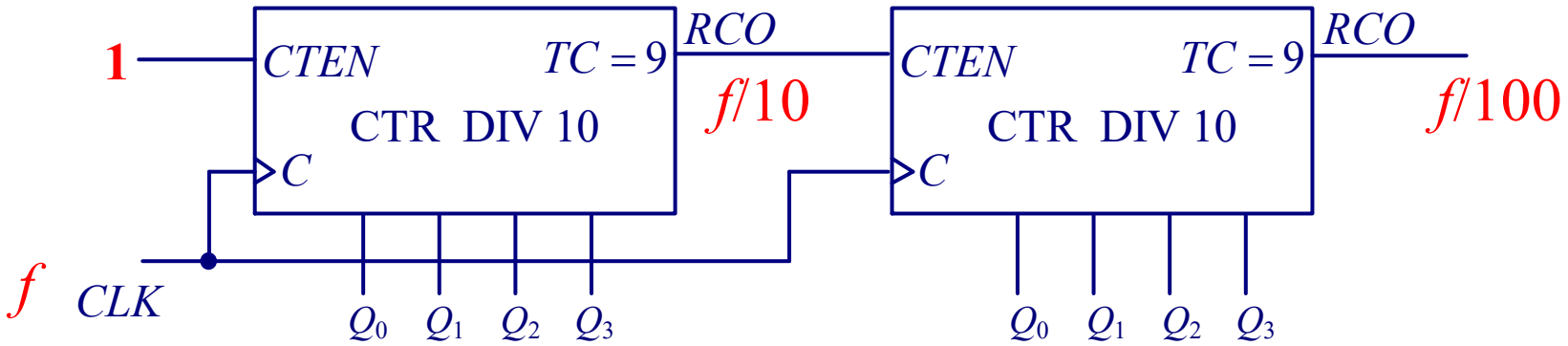
B. 利用集成计数器设计电路

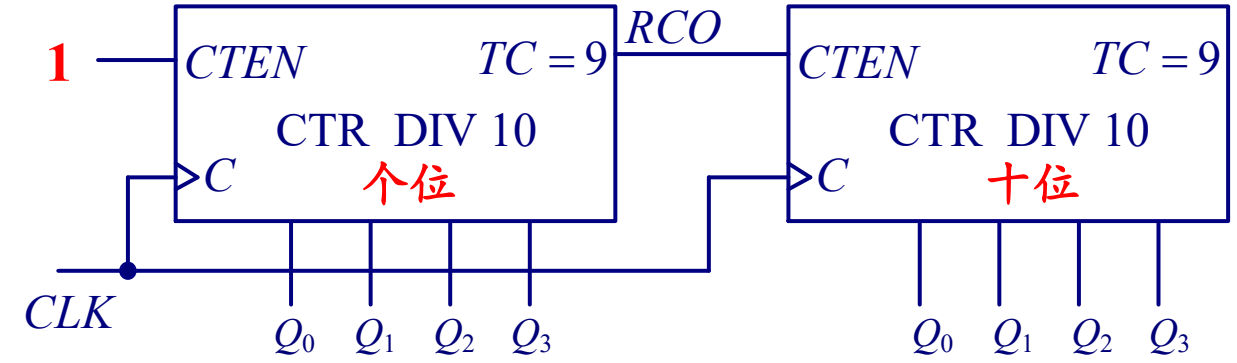
假定已有N进制计数器，
需要得到M进制计数器

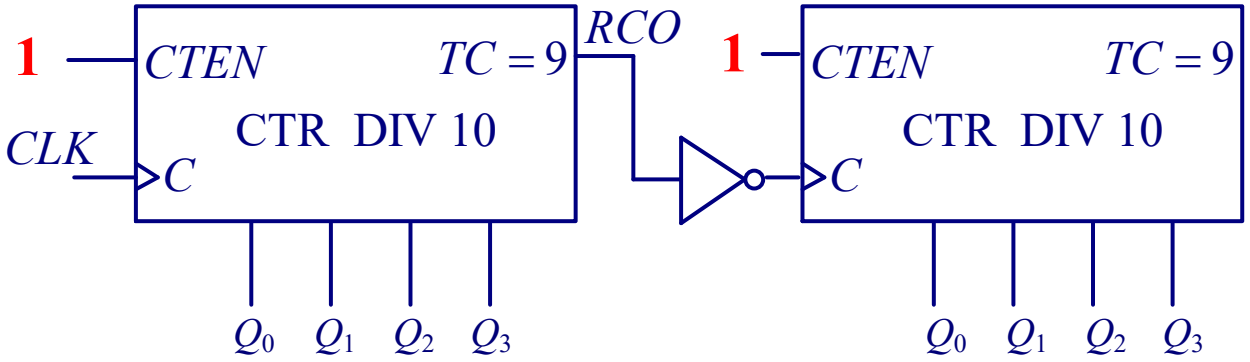
1) M > N 的情况

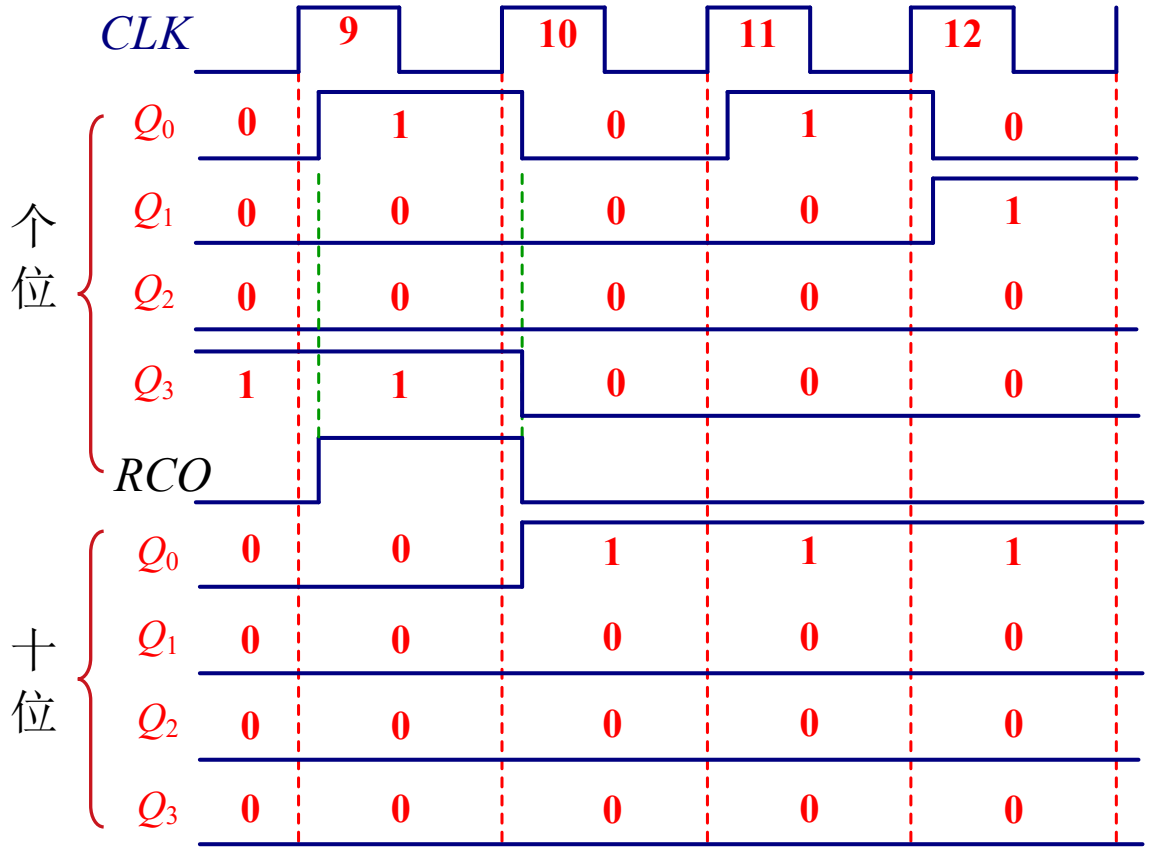
a) 若M=N×N　　cascaded counters



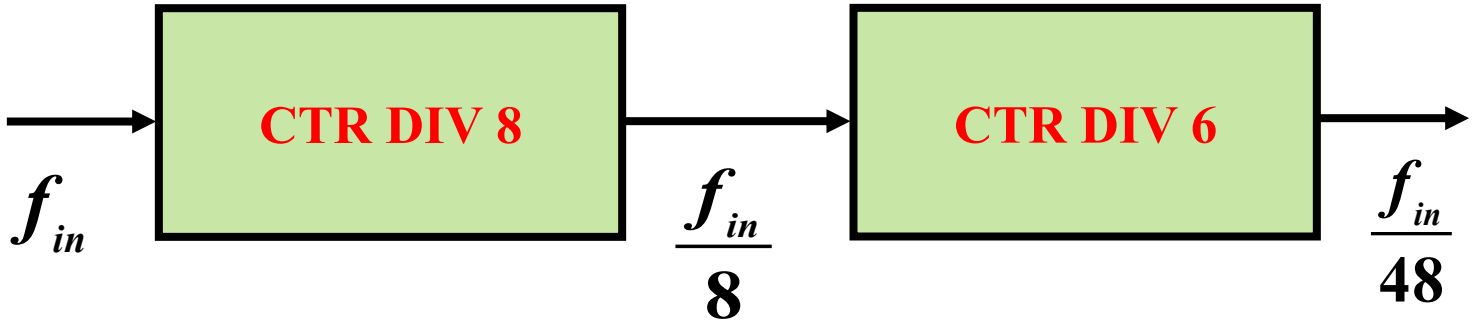



1
CTEN
TC = 9
RCO
CTR DIV 10
C
个位
CLK
Q0 Q1 Q2 Q3
CTEN
TC = 9
CTR DIV 10
C
十位
Q0 Q1 Q2 Q3


1
CTEN
TC = 9
RCO
CLK
CTR DIV 10
C
Q0 Q1 Q2 Q3
1
CTEN
TC = 9
CTR DIV 10
C
Q0 Q1 Q2 Q3


CLK
9
10
11
12
个位
Q0 0 1 0 1 0
Q1 0 0 0 0 1
Q2 0 0 0 0 0
Q3 1 1 0 0 0
RCO
十位
Q0 0 0 1 1 1
Q1 0 0 0 0 0
Q2 0 0 0 0 0
Q3 0 0 0 0 0

b) 若M=$N_1 \times N_2$　　其中N>N_1，N>N_2



c) 当M为大于N的素数时，先将若干片N进制计数器级联成一个大于M的计数器（如N×N进制），然后按照下面M<N的情况实现。

2) M < N 的情况

在N进制计数器的顺序计数过程中，设法使之跳过N-M个状态，就可得到M进制计数器

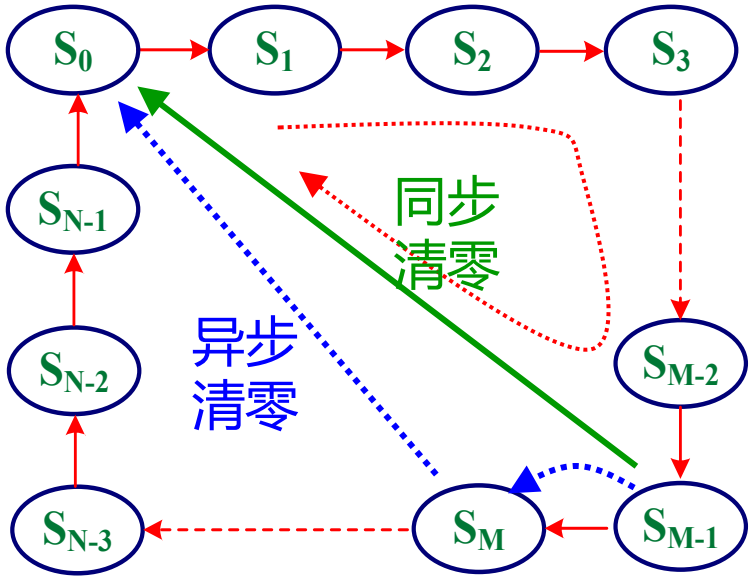
2) M < N 的情况　　Truncated Sequences

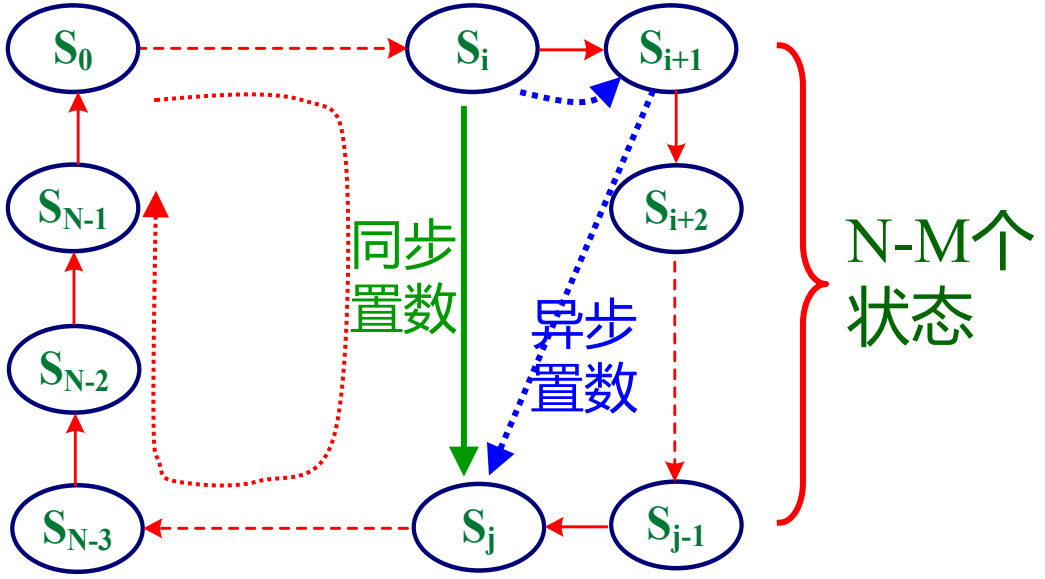
在N进制计数器的顺序计数过程中，设法使之跳过N-M个状态，就可得到M进制计数器

实现跳跃的方式有 清零法 和 置数法

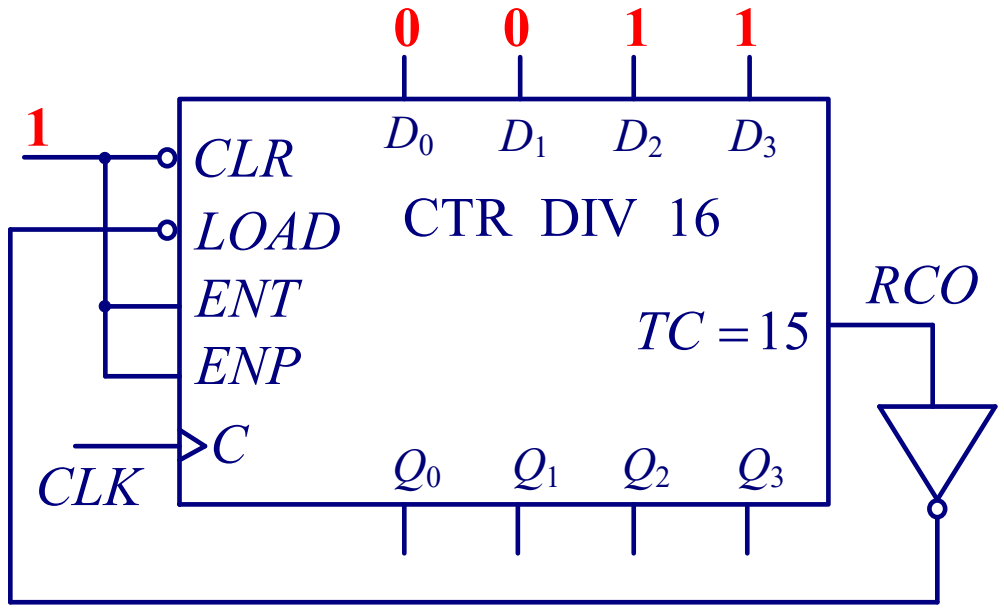
清零法 适用于有清零输入端的计数器

置数法 适用于有预置数功能的计数器





74HC163同步置数



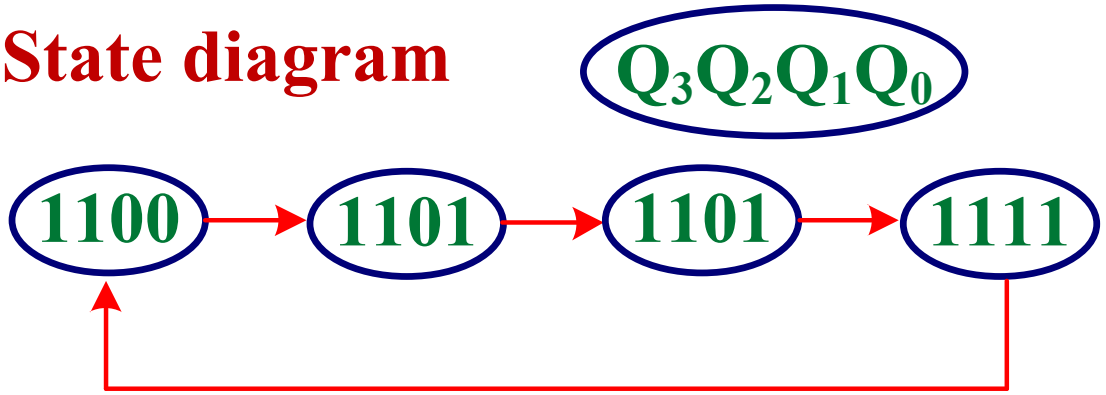
置数 1100

$(1100)_2 = (12)_{10}$

16–12 = 4

Modulo 4 Counter

State diagram

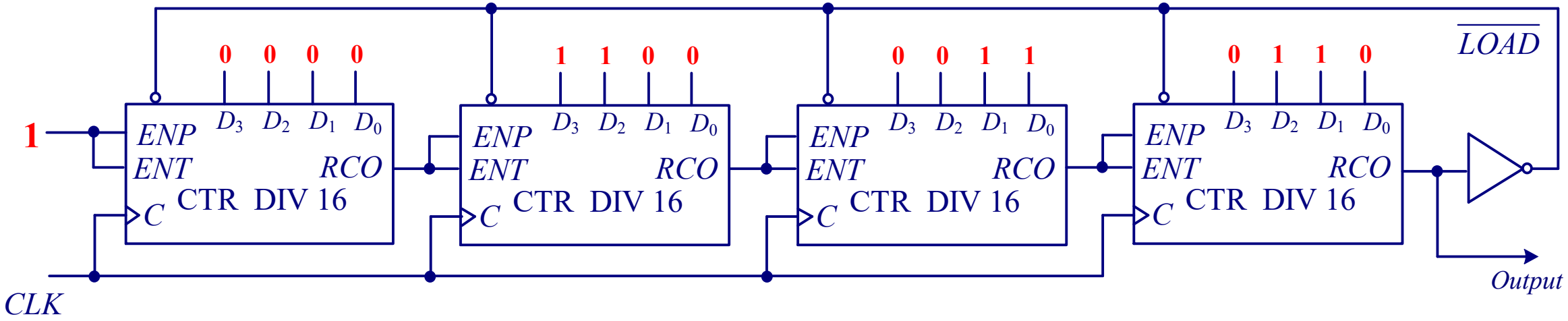


例：用4位二进制同步计数器实现模40000计数器

modulus-16 **a modulus-40,000 counter**

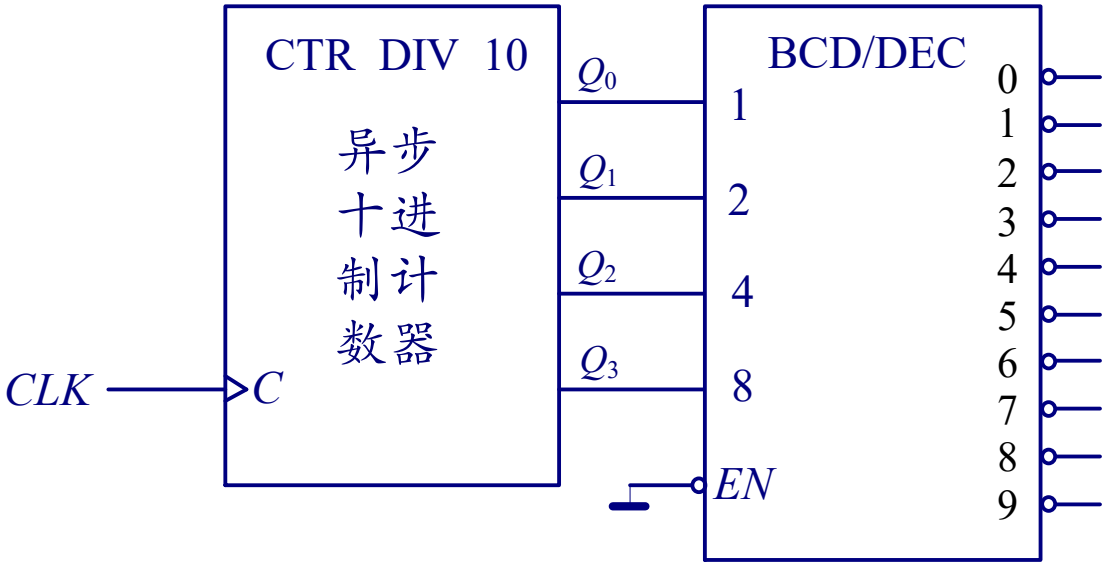
$16\times16\times16\times16 - 40{,}000 = 25{,}536$

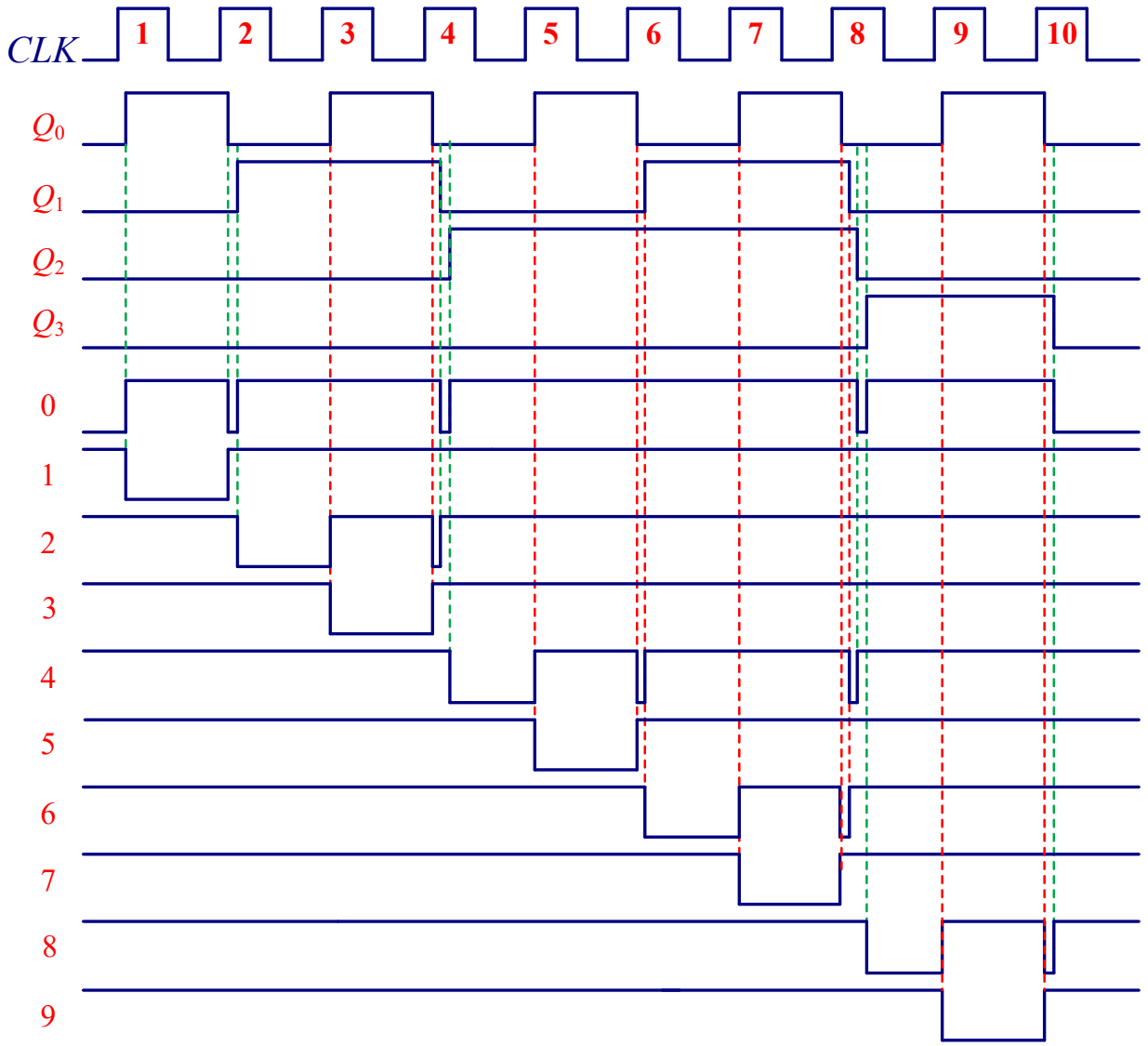
$$(25536)_{10} = (63C0)_{16} = (0110\ 0011\ 1100\ 0000)_2$$

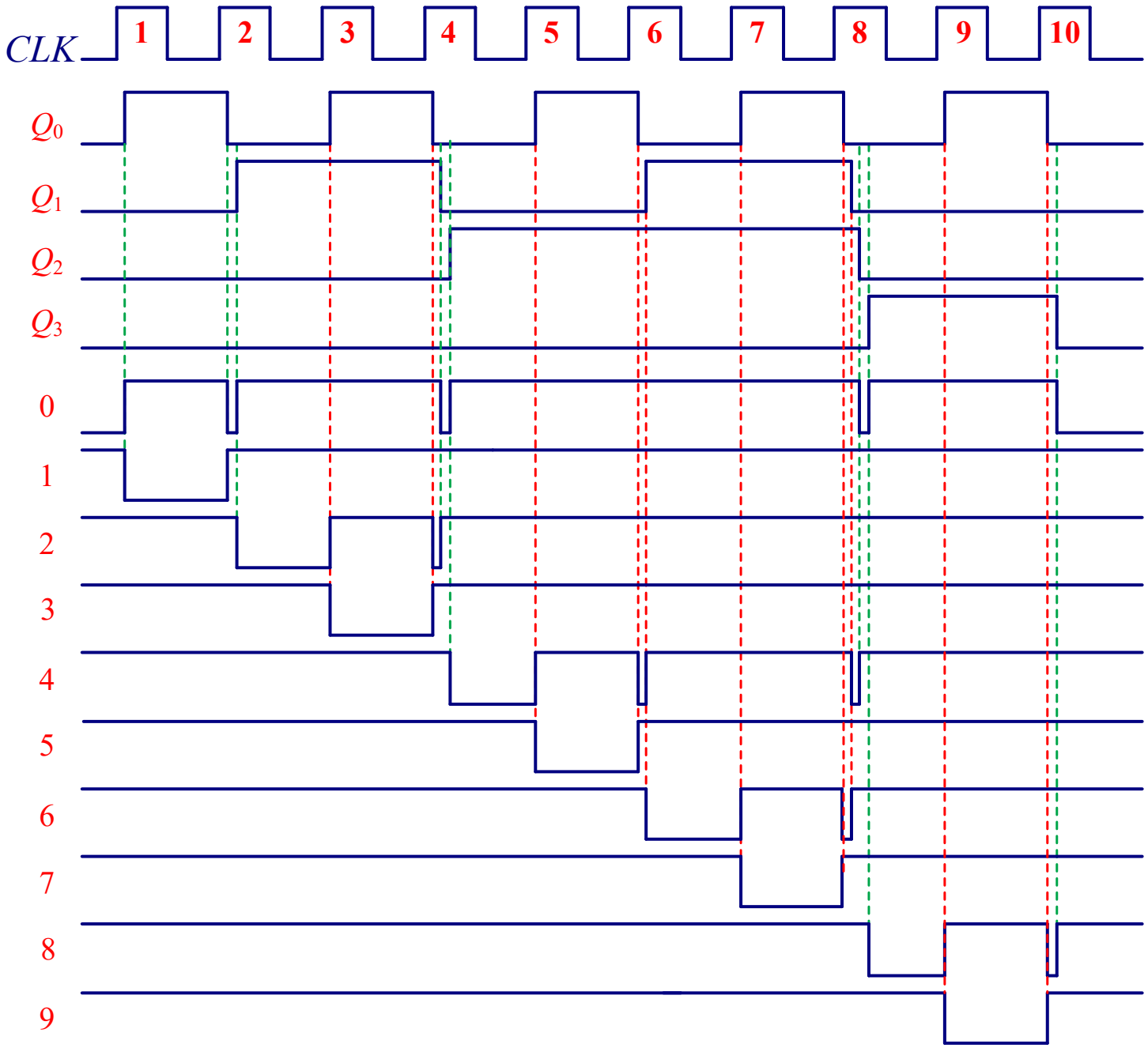


Decoding Glitches

主要讨论计数器译码过程中由于传输延迟引起的译码假信号

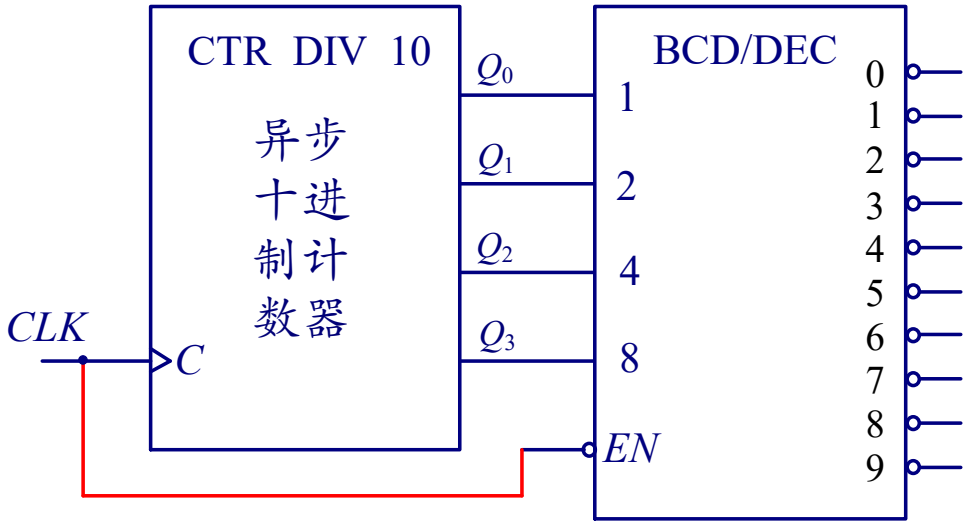


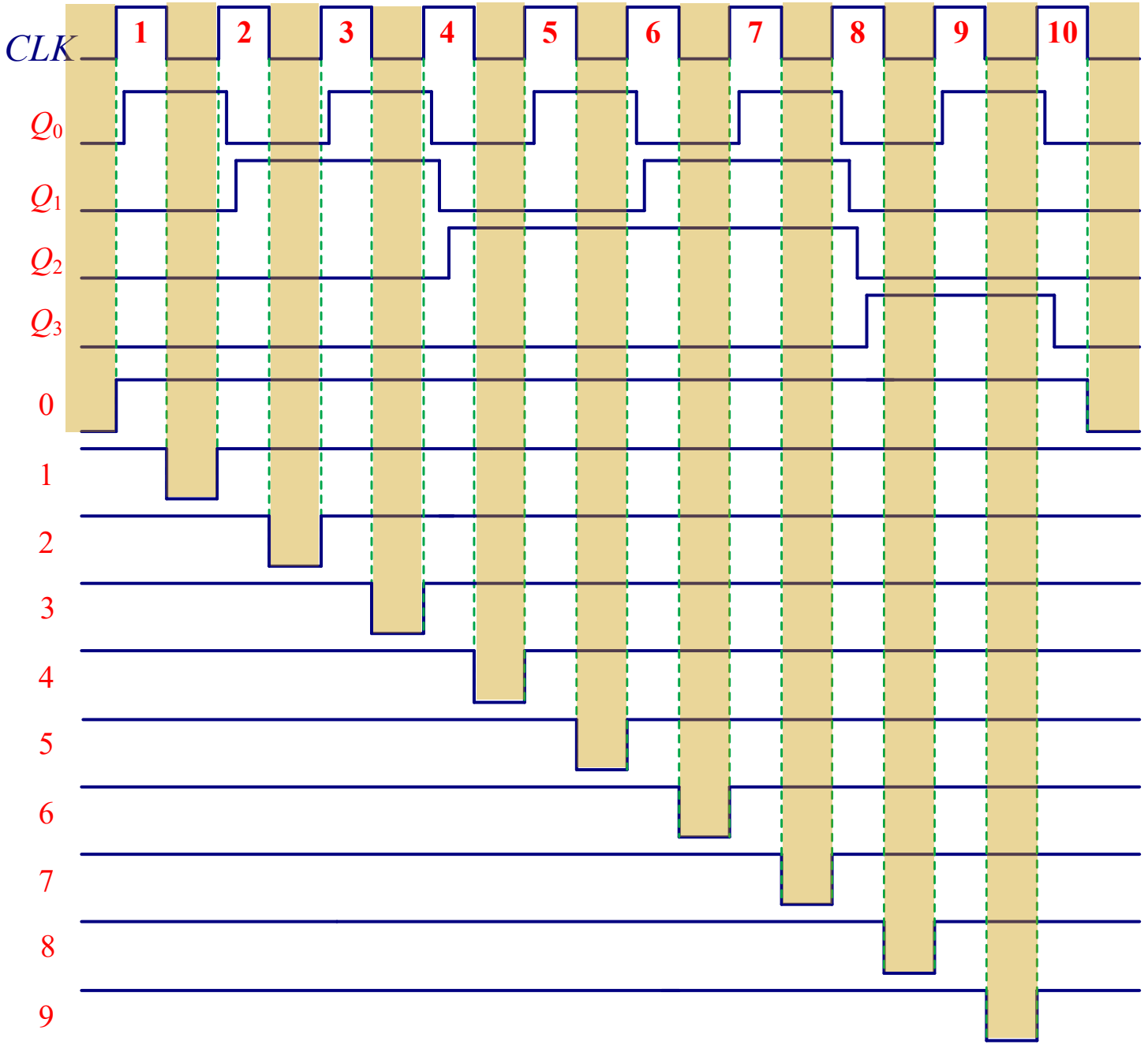




Q3	Q2	Q1	Q0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
0	0	0	0

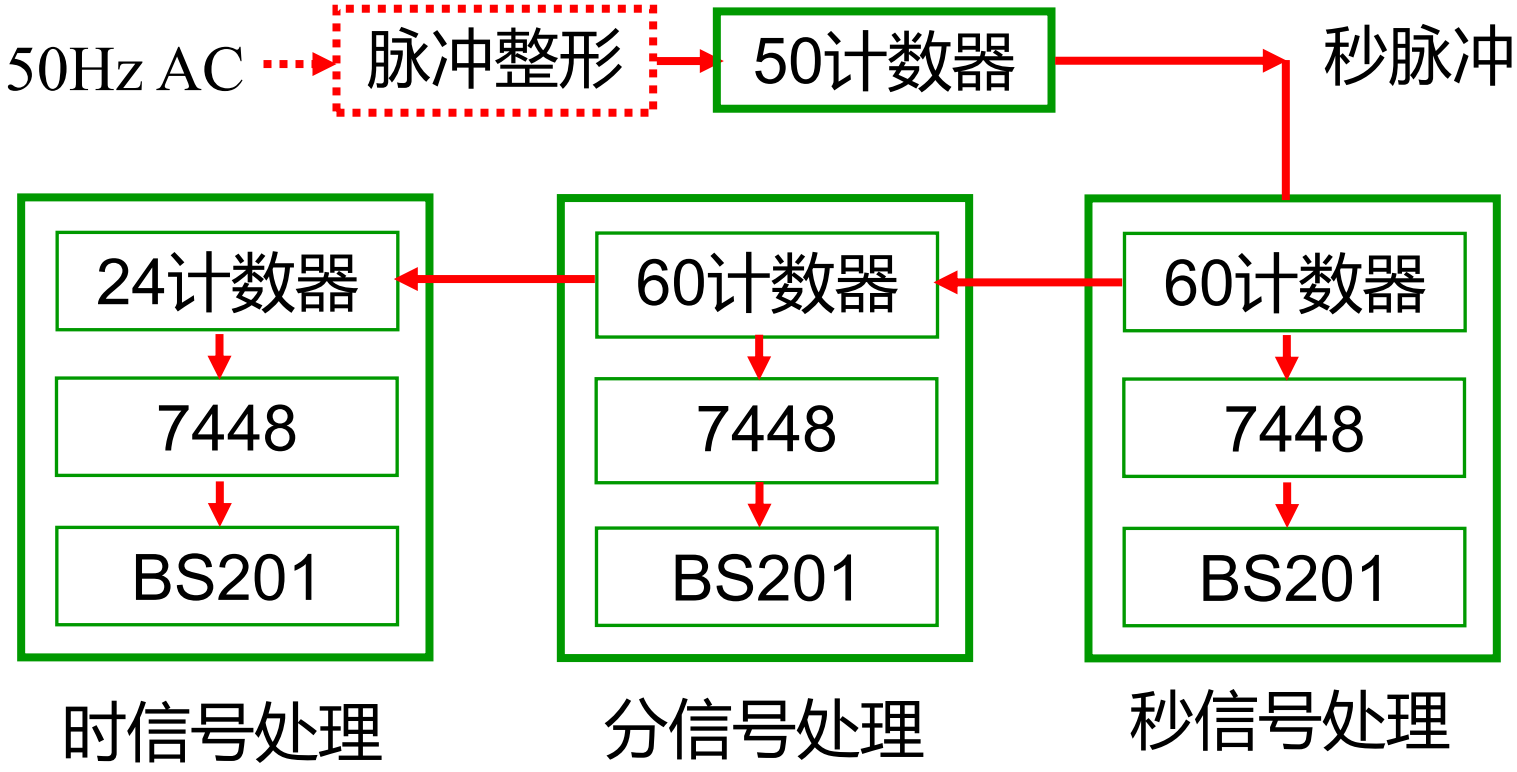
Using strobe signal to eliminate decoding glitches

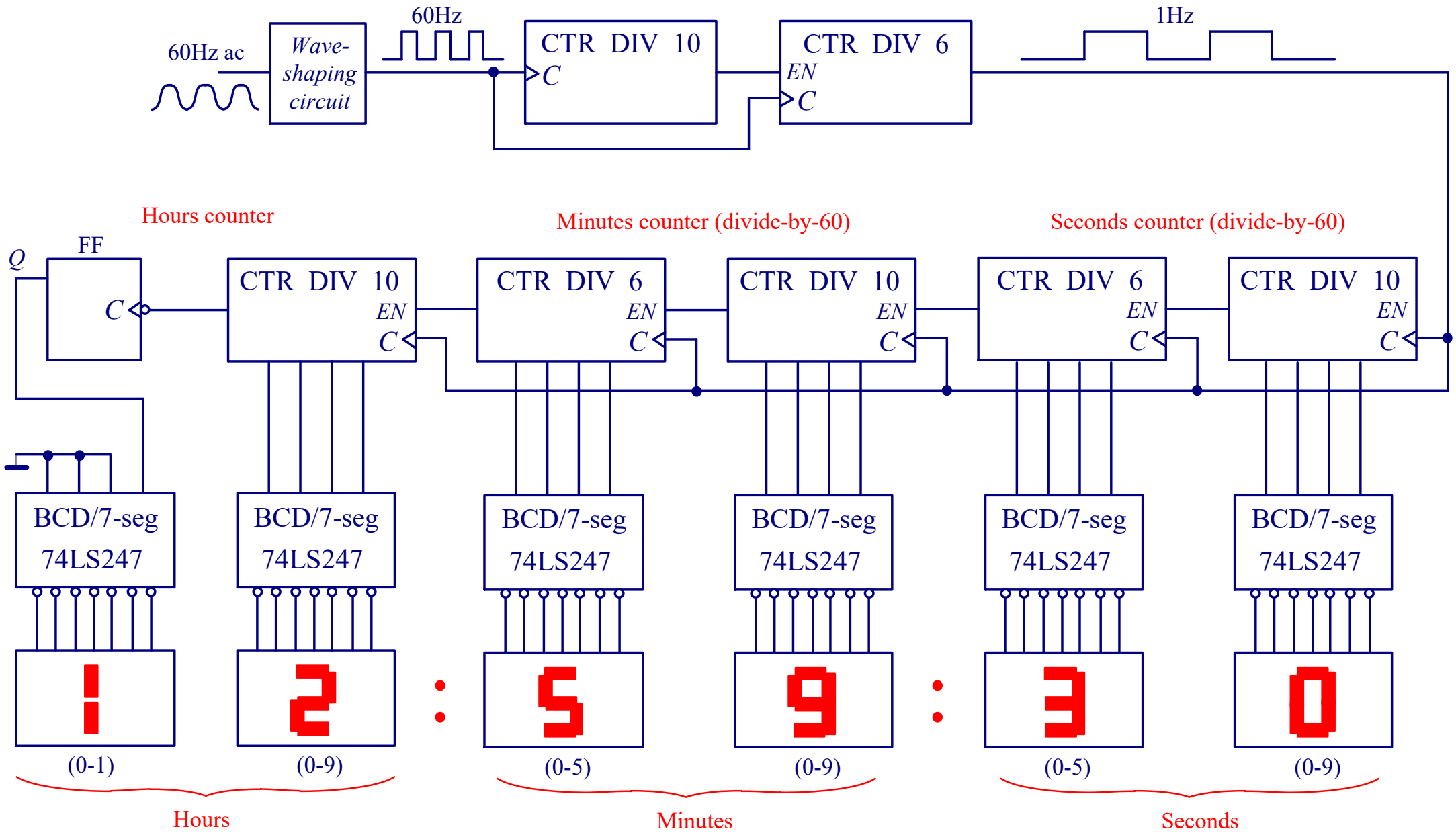




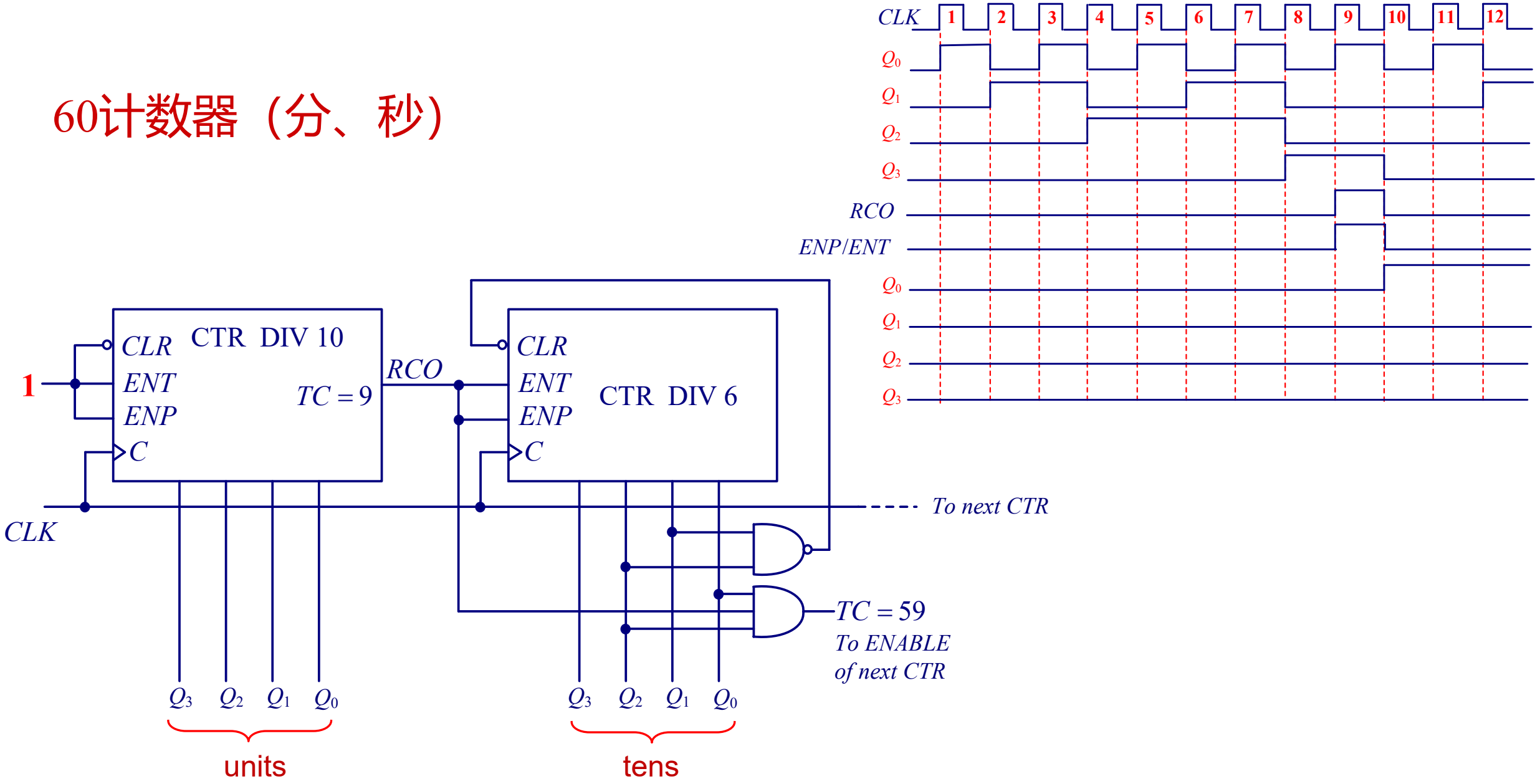
计数器的应用：数字时钟电路，显示00:00:00到23:59:59假设交流电信号(50Hz/60Hz)已用整形器转换成同频率的矩形脉冲信号。

分析：首先需要五十/六十进制计数器将矩形脉冲转换成秒脉冲，计时器需要秒、分、时三个信号处理单元，每个信号处理单元都由计数器，译码器和显示器构成。

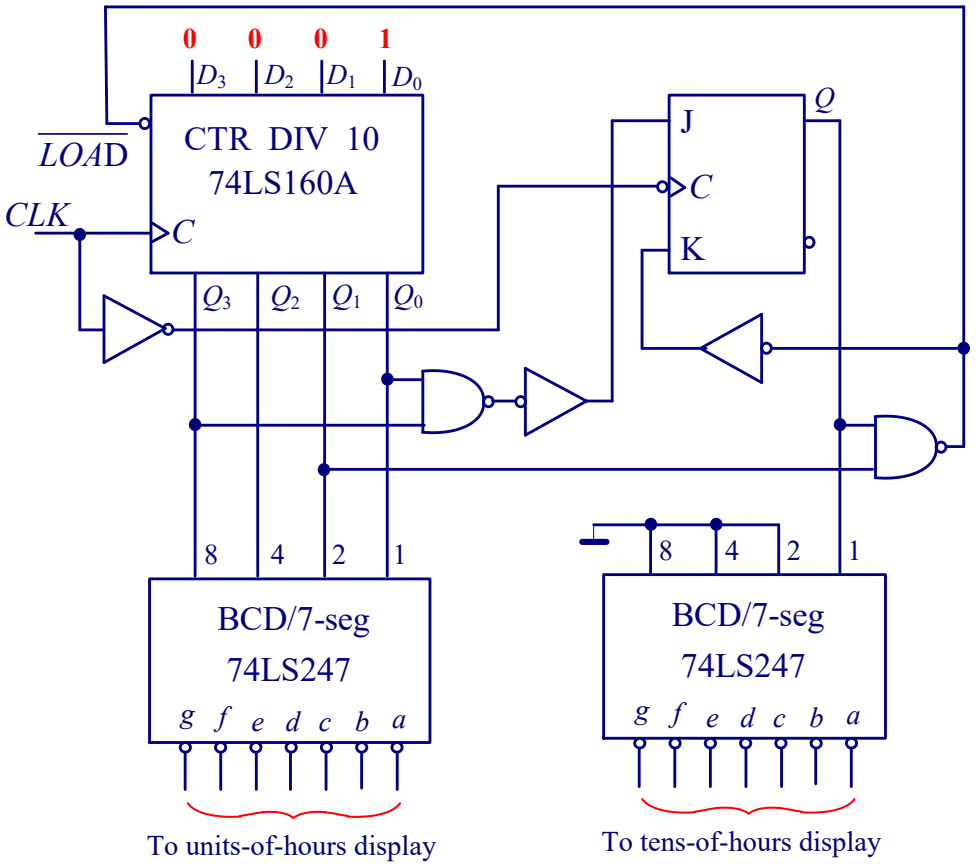


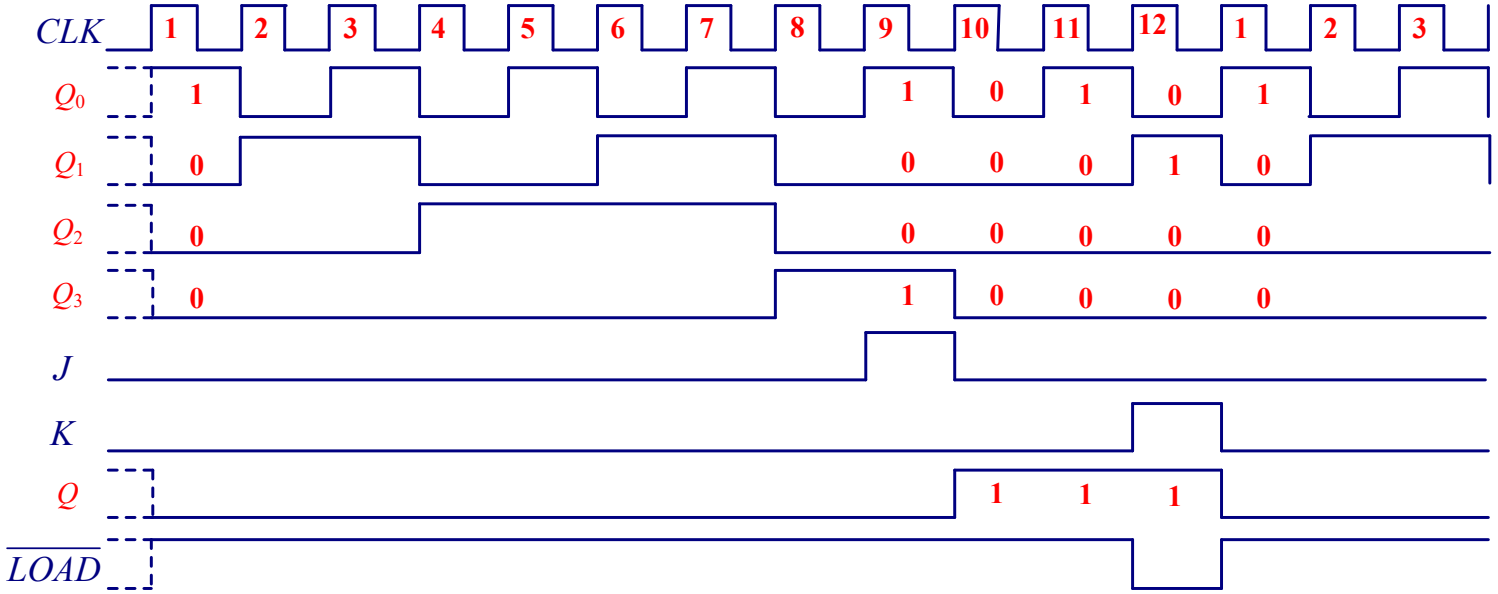

60Hz ac
Wave-shaping circuit
60Hz
CTR DIV 10
C
CTR DIV 6
EN
C
1Hz
Hours counter
Minutes counter (divide-by-60)
Seconds counter (divide-by-60)
Q
FF
C
CTR DIV 10
EN
C
CTR DIV 6
EN
C
CTR DIV 10
EN
C
CTR DIV 6
EN
C
CTR DIV 10
EN
C
BCD/7-seg 74LS247
BCD/7-seg 74LS247
BCD/7-seg 74LS247
BCD/7-seg 74LS247
BCD/7-seg 74LS247
BCD/7-seg 74LS247
1
2
5
9
3
0
(0-1)
(0-9)
(0-5)
(0-9)
(0-5)
(0-9)
Hours
Minutes
Seconds

60计数器（分、秒）


CLK
1
2
3
4
5
6
7
8
9
10
11
12
Q_0
Q_1
Q_2
Q_3
RCO
ENP/ENT
Q_0
Q_1
Q_2
Q_3
1
CLR
ENT
ENP
C
CTR DIV 10
TC = 9
RCO
CLR
ENT
ENP
C
CTR DIV 6
To next CTR
CLK
TC = 59
To ENABLE
of next CTR
Q_3 Q_2 Q_1 Q_0
Q_3 Q_2 Q_1 Q_0
units
tens

12计数器（小时）


0 0 0 1
D_3 D_2 D_1 D_0
CTR DIV 10
74LS160A
$\overline{LOAD}$
CLK
C
Q_3 Q_2 Q_1 Q_0
J
C
K
Q
8 4 2 1
BCD/7-seg
74LS247
g f e d c b a
8 4 2 1
BCD/7-seg
74LS247
g f e d c b a
To units-of-hours display
To tens-of-hours display


CLK
1 2 3 4 5 6 7 8 9 10 11 12 1 2 3
Q_0 1 1 0 1 0 1
Q_1 0 0 0 0 1 0
Q_2 0 0 0 0 0 0
Q_3 0 1 0 0 0 0
J
K
Q 1 1 1
$\overline{LOAD}$

Parallel-to-serial conversion

